

Does Energy Cause Growth? Evidence from Spanish Italian and French energy markets

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Abstract

This research is intended to strengthen the evidence base regarding the interplay between energy production and macroeconomic outcomes. It tests existing hypotheses previously formulated in this field in a cross-country across periods time series analysis. By applying a Bayesian shrinkage approach to a High-Dimensional VAR, this study performs comparisons of energy mixes and their relationship with different macroeconomic outputs in three different countries: France, Italy and Spain. Drawing on more than forty macroeconomic and energy variables, this paper is the first to draw a distinction between economic performances across two separate channels and testing the four different hypotheses on all channels. By extending the BVAR approach with a time-varying estimator (TVP-BVAR) this investigation is finally capable of assessing the evolution of these dynamics, explaining benefits and costs of different mixes and gaining additional insights on the validity of the hypothesis. Overall findings provide evidence of the macroeconomic benefits of maintaining a heterogeneous and domestically diversified energy production mix and the credibility of two of the four hypothesis, depending on the channel of transmission.

I. Introduction

On October 19th, 1973, the Organization of Arab Petroleum Exporting Countries (OAPEC) announced the imposition of an oil embargo against all states that had supported Israel in its war against Arab countries. The effort was coordinated by King Faisal of Saudi Arabia as a response to the increasing aid that Western countries (Canada, Japan, U.K., Portugal and the U.S.) were providing to the Israeli state. As the United States sourced approximately 25% of its oil from these countries, the catastrophic effects of the total embargo were not long to manifest. The inflationary spike during the “first oil shock” reached levels comparable to the Great Depression and the whole economy was largely affected. Although the lifting of the embargo occurred shortly after, in March 1974, its repercussions persisted throughout the following decade.

Since then, the relation between energy prices and macroeconomic aggregates has become a central argument within the macroeconomic literature. Starting from 1978, a substantial body of theoretical literature emerged. The seminal work by [Kraft and Kraft \(1978\)](#) is widely regarded as the foundational work of the Conservative Hypothesis. Using time series analysis from 1947 to 1974, the authors proved a unidirectional Granger causality between economic output and energy production. Economic development and growth are responsible for increases in demand of energy, while policies or external shocks targeting energy prices or supply are not observed to have any impact on economic outputs ([Akarca and Long, 1980](#)).

In stark contrast with this theory, the Growth Hypothesis postulates that energy availability and consumptions are key drivers of economic development acting as irreplaceable complements to capital and labor. Reductions in energetic supply are direct causes behind economic stagnation. In a VAR analysis using data between 1974 to 1990, [Stern \(1993\)](#) demonstrated a statistically significant positive effect of energy consumption on economic output.

An alternative theory was formulated by [Yu and Hwang \(1984\)](#). Using Sims tests on data between 1947 and 1979, they demonstrated the inexistence of any causal relationships between energy consumption and gross national product establishing what became known as the Neutrality Hypothesis ([Cheng and Lai, 1997](#)).

Using modern econometric techniques, more recent studies ([Apergis and Payne, 2009](#); [Algül and Erenel, 2023](#)) have found instead significant evidence supporting the Feedback Hypothesis. Using heterogenous panel cointegration tests ([Pedroni, 1999, 2004](#)) both papers exhibit bidirectional causality between energy consumption and economic output. Additional tests support a short-run unidirectional effect of energy consumption on economic growth and a bidirectional causality between the two variables as the time horizon grows ([Oh and Lee, 2004](#)).

The aim of this paper is to investigate the validity of these hypotheses in a cross-country analysis that employs High-Dimensional VARs and Bayesian VAR computations to illustrate the existing association between energy production mixes and respective economic outputs. Building on the existing literature, this paper not only assesses how

different production mixes interact with economic growth but also extends the research agenda by examining their distinct responses to external shocks. As the current political debate now attributes an increasing importance to the role of energetic independence to shape resilient economic growth, the paper investigates how different production mixes mitigate or amplify the effects of external shocks.

To this end, I will focus my research on three countries that have distinct energy production composition: Italy, Spain and France. Italian long-term reliance on imports is confronted with Spanish path to renewable energy and France’s leading role in European nuclear energy production. Monthly data observations spanning 2007 to 2024 and Impulse Response Function (IRF) analysis will illuminate the association between each energy production source and economic output within each country. The implementation of time-varying Bayesian VARs will further explore the amplification (mitigation) effect that each production mix has with respect to external shocks, thereby generating insights relevant to the evaluation of all four hypotheses.

For this purpose, the dataset at disposition presents three crucial historical events. The sovereign debt crisis and the associated austerity measures allow us to assess whether contractionary monetary and fiscal policies had any significant impact on energy production and consumption. Specifically targeting Spain and Italy, I provide a first assessment on the conservative hypothesis credibility. Covid-19 pandemic offers additional evidence in this direction. Finally, Russian invasion of Ukraine and the subsequent cut in gas importation in 2022, will provide information to understand how energy unavailability and prices may affect macroeconomic output. A comparison of countries’ exposure to this shock are used for a final evaluation of energy dependence.

Expanding the research on this field, the analysis will not treat all macroeconomic aggregates equivalently. Through the imposition of two channels of transmission, and their respective inverses, the paper aims to test the validity of the mentioned hypothesis on the supply and demand sides of the economies: industrial production and household consumption.

The paper is organized as follows: section II. is dedicated to data extraction, manipulation and adjustment procedures, section III. discusses the rationale and the application of the chosen empirical strategies, section IV. presents the main findings and results while section V. concludes the paper.

II. Data

Data were primarily sourced from institutional providers¹, and were subsequently subject to a series of adjustments and an imputation procedure.

¹ *Three accredited channels were used for data not available through European institutional sites: Jing Cynthia Wu website contained the exact values for monthly European Shadow Rate, stox.com was the only available provider having at disposition daily observations of the VSTOXXX volatility index; investing.com is indicated as reliable source from which I extracted the TFF Gas price index at daily frequency; FactSet dataset is used to extract 2015-2025 day ahead spot prices for French energy.*

Variable	Unit	Source	Variable	Unit	Source
Employment Rate	%	Eurostat	10-year bond	% p.a.	Eurostat
Industry Production	index 2015 = 100	Eurostat	VSTOXX volatility	index points	Stoxx.com
Industry Prod. Electricity	index 2015 = 100	Eurostat	Eurostoxx 50	index points	ECB portal
Industry Prod. Manufacture	index 2015 = 100	Eurostat	Loans to NFC	Millions (€)	Eurostat
Retail Sales	index 2015 = 100	Eurostat	Loans to households	Millions (€)	Eurostat
HICP core	index 2015 = 100	Eurostat	Eff. Exchange Rate (REER)	index 2015 = 100	International Monetary Fund
Shadow Rate	% p.a.	WU and XIA (2016)	Producer Price Index	index 2015 = 100	Eurostat
3-month Euribor	% p.a.	ECB portal	TFF natural gas price	€/MWh	Investing.com
Balance of Payments (BOP)	index 2015 = 100	Eurostat			

Table 1: Macroeconomic Variables

o capture a comprehensive picture of macroeconomic trends and of the country-specific production mixes, while leveraging the Bayesian VAR’s ability to handle high-dimensional systems, the final dataset comprises 40 variables, monthly observed for each state between 2007 and 2024. Out of these, 17 accounted for macroeconomic indicators and trends while 23 portray energy production mixes, trades and prices. An addition of two variables was added for the imputation exercise later described.

Table 1 summarizes the most relevant macroeconomic variables that were both available at monthly frequency and for the three countries. They constitute a functional mix of six major macroeconomic areas: labor market, real economic activity, prices and inflation, monetary policy and rates, financial conditions and uncertainty, international trade and exchange.

Table 2 is the result of an extraction exercise mainly carried through the Entso-E platform, a European agency that provides historical data on the monthly production mixes of EU countries. Variables not related to production are those measuring fluctuations in monthly prices and energy import-export balances.

Variable	Unit	Source	Variable	Unit	Source
Hydro Power	GWh	Entso-E	Biomass	GWh	Entso-E
Hydro Renewable	GWh	Entso-E	Other Hydro	GWh	Entso-E
Nuclear	GWh	Entso-E	Total Renewables	GWh	Entso-E
Total Fossil Fuels	GWh	Entso-E	Other Renewables	GWh	Entso-E
Lignite	GWh	Entso-E	Geothermal	GWh	Entso-E
Hard Coal	GWh	Entso-E	Waste	GWh	Entso-E
Natural Gas	GWh	Entso-E	Unclear	GWh	Entso-E
Oil	GWh	Entso-E	Other non-renewable	GWh	Entso-E
Other Fossil	GWh	Entso-E	Import-Export Saldo	GWh	Entso-E/Eurostat
Mixed Fuels	GWh	Entso-E	Wholesale Trade Price	€/MWh	Omie/GME/FactSet
Wind	GWh	Entso-E	Semestral retail (household)	€/KWh	Eurostat
Solar	GWh	Entso-E	Semestral retail (non-HH)	€/KWh	Eurostat
German Wholesale Trade	€/MWh	ISI Franhofer			

Table 2: Energy Variables

The data collected then received four main adjustments. Variables expressed in daily observations (VSTOXX, TFF, Spanish and French wholesale trade prices) were monthly averaged to guarantee frequency conformity. Variables referring to energy flows (prices) were converted into MWh (€/MWh) to ensure unit consistency. All variables expressed as indexes, flows or prices were log-transformed, those expressed as percentages were not.

A seasonal detecting function was applied to all variables for each country, and an X-13 seasonal filter (Findley, 1998) was applied to those presenting cyclical trends over the years (see Appendix A).

After completing these transformations, the absence of French wholesale energy prices for 2007–2015 was addressed through an imputation procedure. Eurostat provides semestral observations for energy retail prices for household and non-household consumers from 2007 onwards. These variables, however, are intrinsically different from wholesale trade prices. While wholesale prices reflect day-ahead spot market dynamics, retail prices are governed by long-term supply agreements between energy providers and buyers. Differences in retail prices are given by the magnitude of the purchase, previous agreements and future allocations, none of which are features of spot market pricing.

To construct a probabilistic series that was not purely linearly imputed, I divided the imputation procedure into two stages. First, I estimated a linear regression model for the available French wholesale energy prices (2015–2024):

$$en_price_{FR,t} = \alpha + \beta_1 en_price_{DE,t} + \beta_2 ret_price_{FR,t} + \beta_3 ret_price_{hh_{FR,t}} + \beta_4 IP_{FR,t} + \beta_5 nuc_{FR,t} + \epsilon_t$$

including German monthly wholesale energy prices, French semestral retail energy prices for household and non-household consumers, French industry production and French nuclear production. I then isolated the stochastic residuals by using these estimates to build a prediction of French energy prices for the known periods. Based on these stochastic residuals, I applied a Kalman Smoother that followed an auto ARIMA model to impute probabilistic residuals for the unknown period. Finally, the imputed observations for French wholesale energy price are the sum of the structural model linear imputation and the imputed stochastic residuals. Once this imputation was completed, the final dataset was ready for estimation.

III. Empirical Strategy

Working with a dataset of more than forty variables necessitates the use of a variable shrinkage method capable of handling high-dimensional VAR. Relying on different assumptions and requirements, these multiple algorithms work differently, provide different outputs and are specifically designed to accommodate certain needs. In identifying the best-suited model for these data, I inspected the patterns within the time series through two main tests.

First, I apply the Augmented Dickey-Fuller test (ADF) (Dickey and Fuller, 1979) to test stationarity on the eighteen variables chosen for the Bayesian VAR specification. In accordance to economic reasoning and by observing the variables evolution over the sample, I assign the “trend test” if the variable is observed to have a deterministic time trend in levels; the “drift test” in the absence of a deterministic trend and with non-zero

mean². Considering the critical values obtained from the Dickey-Fuller distribution, I fail to reject the null hypothesis ($H_0 : \rho = 0$) claiming the presence of a unit root 9 times out of 18, considering a 5% confidence interval. Moreover, some of the series for which the null hypothesis has been rejected are likely to have suffered from several structural breaks within the time period analyzed (energy price, industrial production and retail sales are among these). ADF is not able to accommodate these breaks and spuriously assigns very large negative statistics that make the null hypothesis instantly rejectable. As these episodes are the result of the algorithm conflation of mean-shifting dynamics and mean reversion (Perron, 1989), I consider my sample to be predominantly non-stationary.

Second, I perform a Johansen Cointegration Test (Johansen, 1988, 1991) to identify cointegration relationships in the data. I apply the test to two macroeconomic indicators within the Italian market: retail sales (monthly proxy for consumption) and HICP (harmonized index of consumer prices). The test rejects the first null hypothesis ($H_0 : r = 0$, no cointegration) by producing a trace statistic efficiently clearing Johansen critical values at all levels. The two variables' cointegrating behavior is further demonstrated by the rejection of the second test ($H_0 : r \leq 1$, at most one cointegrating vector) at the 10% critical threshold (results of both tests are reported within Appendix B.).

Evidence from these tests suggests the presence of non-stationary variables and of at least two cointegrated series. Addressing the information loss associated with first-differencing, the inability of frequentist standard models to deal with cointegrated data, and the unbalanced structure arising from changes in the measurement of energy mixes over time, I indicate the Bayesian VAR approach as the best performing model to satisfy the purposes of the analysis (Sims et al., 1990). Thanks to the imposition of the Minnesota Prior defining the tightness of own-lag decay and cross-variables dependencies, the BVAR estimation (Sims et al., 1990) controls for misspecification caused by integrated and cointegrated variables. Furthermore, the model's ability to cope with missing observations, making it preferable to recent frequentist approaches that incorporate prior information through penalisation.

1. Standard Bayesian VAR Estimation, IRFs and FEVD

Implementing Bayesian VARs entails an arbitrary choice of many factors, which include everything from the number of variables, the lag order, the tightness parameters, the scale parameters and the Gibbs sampling repetition. This section covers the rationale behind the choices made on these elements, a short explanation of the mechanism and its conversion into impulse response functions (IRFs), the interpretable outputs later discussed.

Through the imposed Minnesota Prior variable shrinkage, BVAR is an efficient tool controlling for overparameterization of OLS. Nevertheless, it may suffer as well from stability issues emerging from the inclusion of a disproportionate number of variables,

²An additional test is assigned to the export balance as, even if not demeaned, its series reasonably oscillates around zero for all countries

and thus of coefficients to estimate. The practical rule of thumb proposed by Bańbura et al. (2010) to ensure system stability is:

$$\frac{T}{Kp} > 1$$

where T is the number of effective observations, K the number of variables in the system and p the lag order. The seasonal adjustment operated before allows to choose $p < 13$; an information criteria comparison suggested a lag order of $p = 4$ (with max lag = 8, over-restrictive BIC yielded $p = 2$, under-restrictive AIC $p = 8$)³. Other than being aligned with literature, this choice permits the modelling of within-quarter dynamics of monthly observations. As $T = 216$, I included $K = 18$ relevant variables in order to keep the sanity check (3.0) well beyond its minimum mechanical requirement⁴.

The near-random-walk prior that BVAR imposes on each variable's own-lag dynamics allows this model to handle the high degree of collinearity present between variables. In the stacked VAR representation:

$$Y = XB + E, \quad \text{vec}(E) \sim \mathcal{N}(0, \Sigma \otimes I_t)$$

Y is $T \times K$, X the lagged regressor matrix is $T \times (Kp + 1)$, B the coefficient matrix is $Kp + 1 \times K$ and Σ is the $K \times K$ residual covariance matrix. The coefficient on variable k at lag l in the equation for variable j is defined as:

$$B_{jkl} \sim N(\mu_{jkl}, V_{jkl})$$

While the prior mean incorporates the random-walk belief ($= 1$ if $j = k, l = 1$, zero otherwise) the hyperparameters impact B through the prior variance (measure of tightness):

$$V_{jk}(l) = \begin{cases} \left(\frac{\lambda_1}{l^\alpha}\right)^2 & \text{if } j = k \\ \left(\frac{\lambda_1 \lambda_2}{l^\alpha}\right)^2 \cdot \frac{\sigma_j^2}{\sigma_k^2} & \text{if } j \neq k \end{cases}$$

The tuning of hyperparameters depicts the Prior characteristics by defining the tightness applied to each variable relationship. I impose a conservative overall tightness ($\lambda_1 = 0.1$). Benchmarked to a value of $\lambda_1 = 1.0$ that assigns roughly equal weights to Prior and likelihood estimation, this parameter attributes ten times more importance to the Prior compared to each individual coefficient. The variance on variables' own-first lag is therefore $\lambda_1^2 = 0.01$, centered at 1. This imposes coefficients on the first lag that are consistent with the near-unit-root processes found in the data ($\approx 0.8 - 1.2$).

Furthermore, I establish the cross-variable shrinkage λ_2 . By setting $\lambda_2 = 0.05$ the

³As a robustness check a BVAR controlling for 13 lags was also estimated. It contained a total of 14 variables representative of the Cholesky ordering desired (6 macro, 5 energetic, 3 global indicators). Results show no major divergence to the ones of the main specification.

⁴Bańbura et al. (2010) additionally demonstrated that in settings with monthly data and $T \approx 200$, the BVAR reaches practical stability below $K = 20$. They showed how above that threshold the marginal likelihood deteriorates and the forecasting performance collapses, regardless of the Prior tightness

Prior variance on cross-variable coefficients relative to own-variable coefficient is twenty times smaller. This reflects the idea that most variables have weak or indirect effects on each other, meaning that only lags of highly correlated variables are not shrunk, and maximizes marginal likelihood functioning (Giannone et al., 2015). The harmonic lag decay α is set equal to 2. Implying that distant lags have negligible effects, the quadratic decay ($\alpha = 2$) shrinks the standard deviation on coefficient lags at the rate $1/l^\alpha = 1/l^2$. Following standard literature (Litterman, 1986), this decaying rate protects against spurious significance found due to casual patterns in data rather than economic meaningful coefficients, something common in lasso and frequentist approaches.

In systems with variables expressed in different scales like this one, the imposition of scale parameters σ_j^2 and σ_k^2 , is a fundamental passage to assure that the Prior is equally tight in terms of explained variance, rather than raw coefficient magnitude (Kadiyala and Karlsson, 1997). They are computed from AR(4) residual differences on first-differenced series, and their ratio enters directly in the Prior coefficient's variance estimation. In the following passage, through the diffuse constant Prior I set the variance attributed to the intercept to one million ($var = 1e06$). Aligned with literature standards, this parameter fixes no long-run mean preserving trend, letting the data to freely determine it.

Lastly, as Σ is unknown, the closed-form computation of B is not feasible. For this reason, I implement 12,000 Gibbs sampling iterations (discarding the first 6,000 as burn-in) that produce two conditional posteriors to complete the parametrization:

$$p(B|Y, \Sigma) \propto p(Y|B, \Sigma)p(B)$$

the Normal posterior,

$$p(\Sigma|Y, B) \propto p(Y|B, \Sigma)p(\Sigma)$$

defined as the inverse-Wishart posterior. Each of these remaining 6,000 joint draws $\{B^{(s)}, \Sigma^{(s)}\}_{s=1}^{6000}$ represent a complete parametrization of the system from which IRF and FEVD are derived.

Impulse Response Functions (IRFs) are extracted from the parametrization of the BVAR algorithm and needed to analyze variables' reaction to correlated variables shocks. Through a Cholesky decomposition I can define the ordering and the nature of each variable's shock-response. Defining structural shocks as:

$$\varepsilon_t = P^{-1}e_t$$

the structure of the lower-triangular P^{-1} enforces the casual ordering. Variables at the top of this ordering are exogenous to all other shocks in the system and respond only to their own structural innovations. Variables at the lowest rank of the Cholesky chain are exogenous to none, they are contemporaneously hit by their shocks and by the stream of shocks of the whole chain. The structural IRF at horizon h for one standard deviation shock to variable j on variable i , and differently estimated for each draw s , can

be therefore determined as:

$$IRF_{i,j}^{(s)}(h) = [\Phi_h P^{(s)}]_{i,j}$$

where the matrices Φ_h are obtained from the Vector Moving Average of infinite order ($VMA(\infty)$) representation and recursively computed from the VAR companion matrix.

The Forecast Error Variance Decomposition (FEVD) is a tool used to decompose the total unpredictability of a variable. It divides this unpredictability into percentage contributions of each structural shock within the system at different horizon, highlighting the short-term contemporaneous drivers and the long-term persistent ones. The contribution of shock j to the forecast error variance of variable i at horizon h is:

$$\omega_{i,j}(h) = \frac{\sum_{l=0}^{h-1} (\Theta_{l,ik})^2}{\sum_{l=0}^{h-1} \sum_{k=1}^K (\Theta_{l,ik})^2}$$

where $\sum_{l=0}^{h-1} \Theta_{s,ij}^2$ is h-step ahead forecast error:

$$y_{i,t+h} - E_t y_{i,t+h} = \sum_{l=0}^{h-1} \Theta_{l,i} \cdot \varepsilon_{t+h-l}$$

and $\sum_{l=0}^{h-1} \sum_{k=1}^K (\Theta_{l,ik})^2$ is its variance (the $MSE_i(h)$).

2. Time-Varying Bayesian VAR (TVP-BVAR) Estimation and IRFs

Introduced by (Primiceri, 2005) this Bayesian Vector Autoregression expansion addresses one of the most criticized aspects of fixed coefficient BVAR. In estimations over long time periods, standard BVAR intrinsically confound the transmission mechanisms with changes in shock sizes (Sims, 2001). By allowing for time-varying coefficients, time-varying contemporaneous relationships and time-varying stochastic volatilities, the TVP-BVAR full specification accounts for the evolutionary dynamics contained in the data (Cogley and Sargent, 2005). The estimation equation in the state-space form:

$$y_t = X_t' \beta_t + A_t^{-1} \Lambda_t^{1/2} \varepsilon_t$$

Through the time-indexed parametrization of the coefficients β_t (being a $Kp \times 1$ stacked vector), the $K \times K$ lower triangular Cholesky matrices encoding contemporaneous relationship (A_t) and the $K \times K$ diagonal matrix for shock variances (Λ_t), this specification allows for changing estimates at each time-period. Rather than fixed scalars, these elements are now time-indexed sequences of parameter estimates. β_t , for example, is a vector containing all VAR slope coefficients at time t . The second novel characteristic is that each of these vectors follow a random walk evolution in time. By not imposing a mean-reversing mechanism, coefficients, relationships and shock variances are allowed to permanently drift. The extent to which these parameters are permitted to evolve is

governed by the hyperparameters Q , S and $\sigma_{\xi_i}^2$, themselves estimated from the data with hyperpriors following [Primiceri \(2005\)](#) original calibration.

The TVP-BVAR full parametrization $p(\beta_{1:T}, \alpha_{1:T}, h_{1:T}, Q, S, \sigma_{\xi_i}^2 | Y)$ is divided in four blocks:

- i. Using Kalman filter, the first forward pass is used to recursively compute β between β_0 to β_T . The simulation smoother is then used to compute the backward pass, which starts from β_T and draws backward for $t = T - 1, \dots, 0$.
- ii. After transforming the system to isolate α_t , the free elements of the lower triangular A_t , the two steps are repeated in the second block to draw coefficients of $\alpha_{1:T}$.
- iii. Given coefficient $\beta_{1:T}$ and relationships $\alpha_{1:T}$, each volatility path $h_{i,t}$ (diagonals of the Λ_t matrix) is recovered from the demeaned residuals $e_t = A_t(y_t - X_t' \beta_t)$ where $e_{i,t}^2 = \exp(h_{i,t}) \cdot \varepsilon_{i,t}^2$.
- iv. In the fourth block hyperparameters are applied to the coefficients ⁵.

Given the larger number of parameters to estimate, and retaining the same prior specification, I decrease the variables to $K = 5$. Unlike the fixed-coefficient specification, I also reduce the number of Gibbs iterations to 3,000 (1,000 of which burn-ins). The algorithm product $\{\beta_{1:T}^{(s)}, \alpha_{1:T}^{(s)}, h_{1:T}^{(s)}\}_{s=1}^{2000}$ is then used to compute impulse response functions. TVP-IRFs are estimated accounting for time-specificity. For each draw s , the previously constant structural impact matrix, is now defined at each calendar date t as:

$$P_t^{(s)} = A_t^{(s)-1} \Lambda_t^{(s)1/2}$$

The consequent Impulse Response Function at horizon h for variable i given one standard deviation shock in variable j occurred at time t , and differently estimated for each draw s , is defined as:

$$IRF_{i,j}^{(s)}(h, t) = \Phi_{h, \beta_t}^{(s)} \cdot [P_t^{(s)}]_{ij}$$

where $\Phi_{h, \beta_t}^{(s)}$ are the VMA coefficient matrices at horizon h .

IV. Results

Presenting the results obtained from the implementation of the described models, this section will be divided in three parts. After an evaluation of the Impulse Response Functions obtained by standard BVAR estimation, the second part will concentrate on the Forecast Error Variance Decomposition to determine percentage contributions. The last part is dedicated to the investigation of the coefficient evolution over time, and through shocks, within the three different countries.

⁵For full derivation and algorithm description please refer to [Primiceri \(2005\)](#); [Koop and Korobilis \(2009\)](#).

1. Fixed-Coefficient BVARs Impulse Response Functions

The standard Bayesian VAR estimation was conducted on a sample composed by eighteen variables, four lags and the previously described hyperparameters⁶. To test the initial hypothesis, I designed two economically meaningful transmission channels that targeted both the production and the consumption sides of the economy. By reversing the Cholesky ordering, and thus altering the assumed exogeneity structure, I arrive at a total of four transmission channels.

The production direct channel, assessing how energy mixes influence final production outcomes; the consumption direct channel, which measures the effects of energy mixes shocks on subsequent consumption; the production inverse channel determining the impact of production shocks on energy mixes; and finally, the consumption inverse channel, quantifying consequences of consumption shifts on energy mixes.

To this end, out of the 18 chosen variables, I selected 5 global variables: TFF gas price, Euro STOXX index, VSTOXX volatility index, European Shadow Rate and 3-months Euribor; 5 energy-mix variables: total fossil, total renewable, nuclear, import-export balance and energy price; 8 macroeconomic country-specific indicators: industry production, retail sales, core harmonized inflation, unemployment, 10-year bond yields, producer price index, real effective exchange rate and loans to non financial corporations. Assuring control for global common shifts, I then build the two Cholesky orders on country specific variables.

1.1. Production Direct Channel

In the first transmission channel (the production direct channel), I assume that energy mixes are exogenous and can only be affected by their own shocks. Descending through the chain I assume shifts in energy production affect the energy price, contemporaneously impacted by its own shocks as well, which then influence the producer price index. The PPI response reflects these shocks and has some effect on the final industrial production. In this Cholesky hierarchical ordering, industrial production is exogenous to none, it is contemporaneously affected by its own shocks and the PPI one, where this latter is composed of the shocks of previous variables. Chronologically, industrial production is the last affected variable: its response to energy production mixes is delayed by the previous passages in the chain and observed only months after. Figure 1 displays energy mixes shocks median effects on industrial production on a 24-month horizon, disaggregated for country, and with reported 68% credible bands.

⁶*Geweke (1982) convergence diagnostics indicated parameter failure rates centered around 5%, consistent with expected Type I error, confirming MCMC convergence.*

Variable	FR	ES	IT
Renew total	14.29%	13.75%	12.41%
Fossil Total	24.11%	14.07%	7.57%
Saldo	1,664.48 GWh	472.78 GWh	734.62 GWh

Table 3: Standard Deviations in Shocking Variable

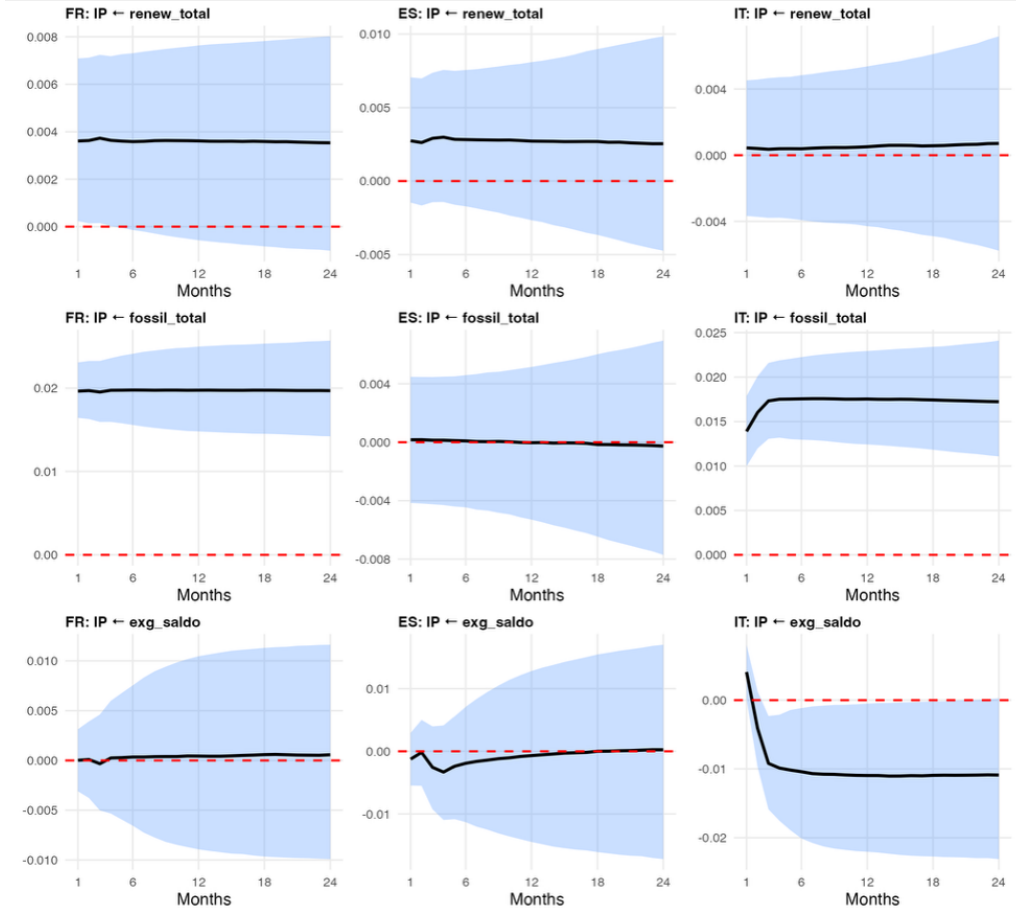


Figure 1: Channel 1 - Energy Mix Shocks on Industrial Production

As expectable, over the 14-year sample starting in 2007, renewable energy shocks fail to have a significant impact on all countries. Although the effects are statistically insignificant, renewable energy shocks are more pronounced in Spain and France than in Italy, whose reliance on fossil sources is highlighted in the second column of the grid. Although one standard deviation shock has same effects in Italy and France (≈ 2 p.p. increase in industrial production), this increment is caused by a 7.57% and a 24.11% increase in fossil production respectively. Italian dependence is evident and furthermore proven by the significance in export balance, France's greater resource diversification is illustrated in Figure 3 showing positive quasi-significant impacts of nuclear energy on industrial production.

Italian sudden increase in energy imports has an initial positive effect on industry

production but reverses its effect within three months. Expansions of imports should occur in the aftermath of positive economic performance, but this demand-driven increase appears unable to offset with higher costs (external markets and transportation) and result in an overall decrease in industrial production. I also compute a full-chain decomposition studying the IRFs of each variable on its next one in the Cholesky order imposed, see Appendix Figure C.1.

1.2. Consumption Direct Channel

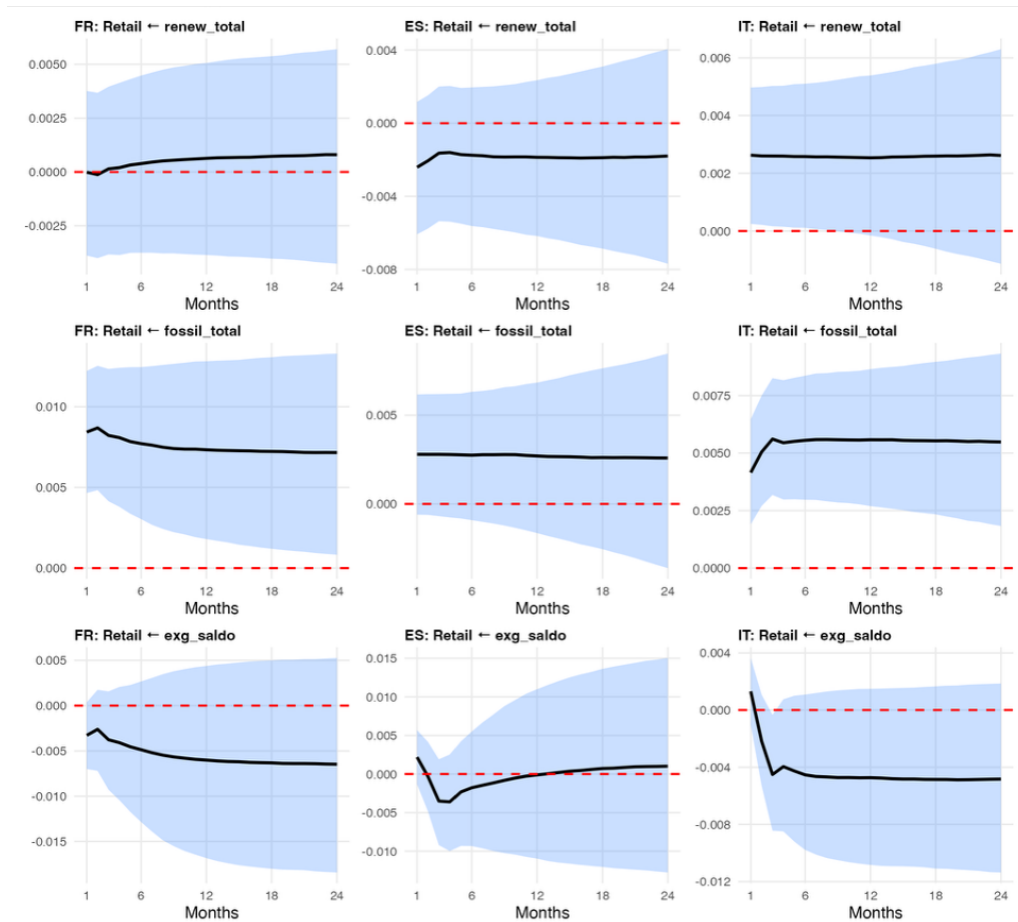


Figure 2: Channel 2 - Energy Mix Shocks on Retail Sales

The second direct channel explored in this section is the consumption channel. Moving to the demand side, I created a Cholesky ordering designed to transmit energy shocks on household consumption. The first two elements of the chain are the same, but in this setting, I observe the impact of energy prices on consumer inflation (HICP) rather than on producer prices. The last chronologically affected variable is retail sales, functioning as a proxy of consumption, it is exogenous to none and contemporaneously affected by its own shocks and the ones compounded through the chain and transmitted by HICP.

Figure 2 depicts the effects of one energy mix standard deviation on retail sales. The IRFs produced within this channel are generally less statistically significant. Italy's de-

pendence on fossil fuel sources and the negative effect of increases in imports is well shown in these responses as well. As reported in the full-chain decomposition (Appendix Figure C.2), Italian inflation is the one suffering the most from increases in energy prices (Conflitti and Luciani, 2019), and these hikes in inflation negatively affect the retail consumption. Combined with Italy’s heavy reliance on imported fossil fuels, these transmission linkages suggest that a meaningful share of Italy’s macroeconomic vulnerability is attributable to its energy supply chain.

French energy production from non-renewable sources is a determinant explainer in this case as well, while nuclear capacity remains a less relevant one. Within the sample, nuclear energy in France has suffered from numerous changes in trends. One of the most significant occurred in 2022, when, in the days preceding the Russian invasion of Ukraine, France initiated a reactor maintenance programme that took several nuclear plants offline. With a solid base provided by nuclear energy, France’s reliance on fossil fuels is likely activated during periods of economic expansion both for demand and supply, explaining its strong positive relation with both.

Spanish estimates remain insignificant. Its uniqueness on avoiding reliance on fossil fuels and energy imports compared to its neighbors is evident but the negative effect of renewable energies on consumption is counter intuitive. Having adjusted for seasonality, my hypothesis is related to government grants for renewable energy production. Allocated as means of economic resurrection at times of poor economic performance, these subsidies were then interrupted once the recovery was complete, explaining the negative, yet insignificant, relation with retail sales.

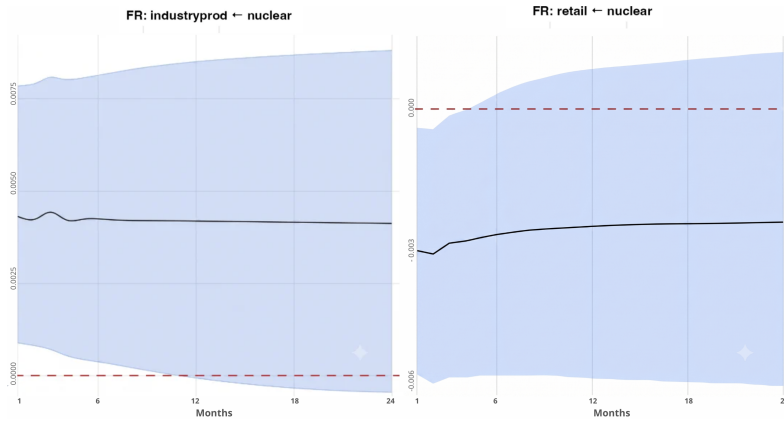


Figure 3: France: Nuclear Generation Shocks

1.3. Production Inverse Channel

The analysis proceeds with the evaluation of the inverse channels. Testing Hypothesis 1, which describes the unidirectional impact of economic aggregates on energy production mixes, I firstly compute the impulse response functions of energy mixes when the shocks to the supply side of the economy are applied. By fully inverting the first transmission channel, I build a Cholesky hierarchical ordering when industry production is exogenous to producer price indexes, energy prices and production mixes. In this setting,

shocks of industrial production depend on decisions unrelated to energy prices (government policies, business cyclical fluctuations or demand contractions) and hit the producer prices. Furthermore, PPI impacts the quantity of energy demanded, thus its prices and production mixes. Changes in energy supply are therefore endogenous, chronologically lagging, and depend entirely on the chained events starting from decreases in industrial production.

Figure 4 shows how one standard deviation in industry production (5.17% in France, 6.53% in Spain, 6.08% in Italy) impact the energy mixes in a 24-month horizon.

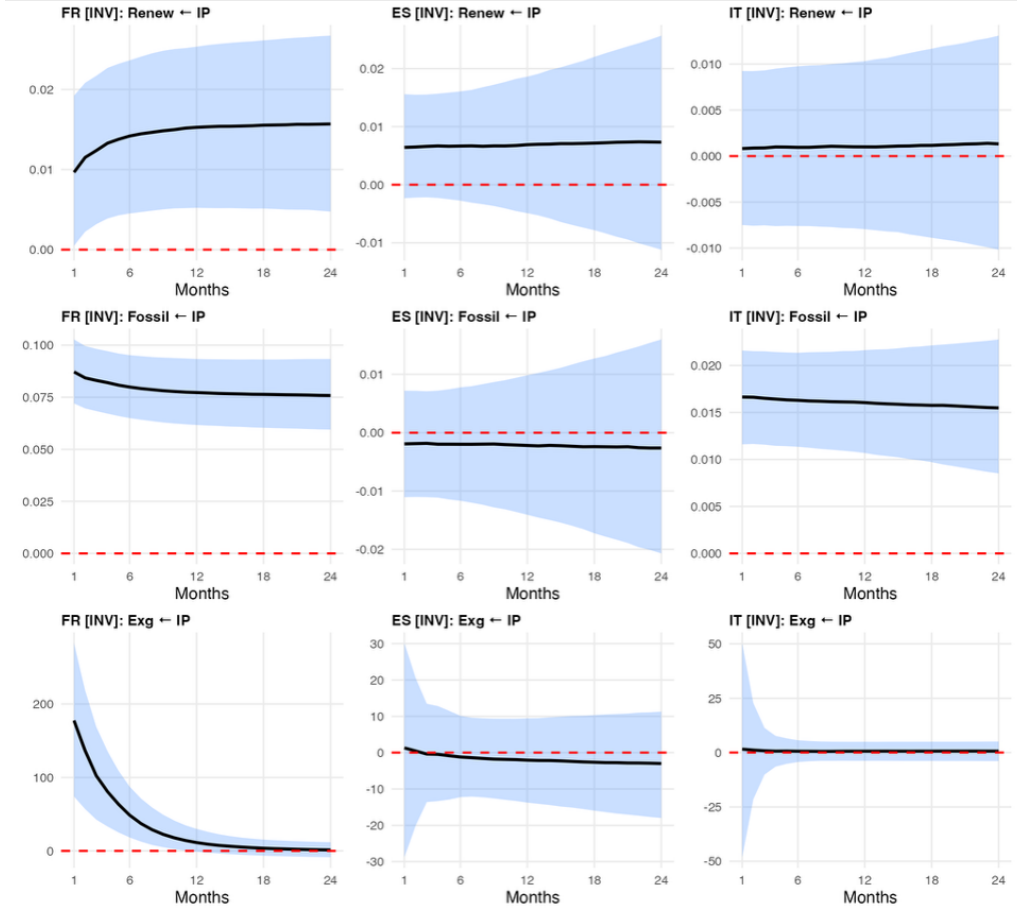


Figure 4: Inverted - Direct Effect of IP Shock on Energy System

Patterns already documented in the previous analysis find reasonable support in these IRFs, pointing towards the credibility of Hypothesis 3 on feedback. Both French and Italian reliance on fossil production sources is further recorded in this inverse channel, while Spanish diversified production model is implied by the insignificance of these results as before. The positive effect of industrial production shocks on PPI and of this latter on energy prices (see Appendix Figure C.3 for decomposition), is absorbed by the negative relation linking prices and renewables for Spain (already documented in Appendix Figure C.2) but not for Italy and France. Emerging as another important advantage deriving from independency, flexibility in the sources of production is translated in positive responses in renewable energy production at times of economic expansion, where the

deflated energy price attenuate for the costs. Italy that instead relies by a large share on fossil fuels but does not produce them, faces higher costs at times of economic expansions, limiting its within borders benefits and sharing these premiums with fuel exporting countries.

1.4. Consumption Inverse Channel

In this last analysis I revert the Cholesky ordering for the second transmission channel. The assumption that energy mixes had direct effect on inflation and consumption is inverted and retail sales are now at the top of the Cholesky structure. Exogenous to all, this consumption proxy is believed to have positive effects on inflation, which therefore transfers to the prices of energy and to the energy production mixes. Chronologically last and exogenous to none, the production mixes now represent the responses to increases in demand, which presumably have some impact on energy demanded.

Figure 5 displays this relation on a 24-month horizon while Appendix Figure C.4 the full chain decomposition. One standard deviation shock in retail sales corresponds to an average increase of 5.64 % in France, 5.39 % in Spain and 3.49 % in Italy.

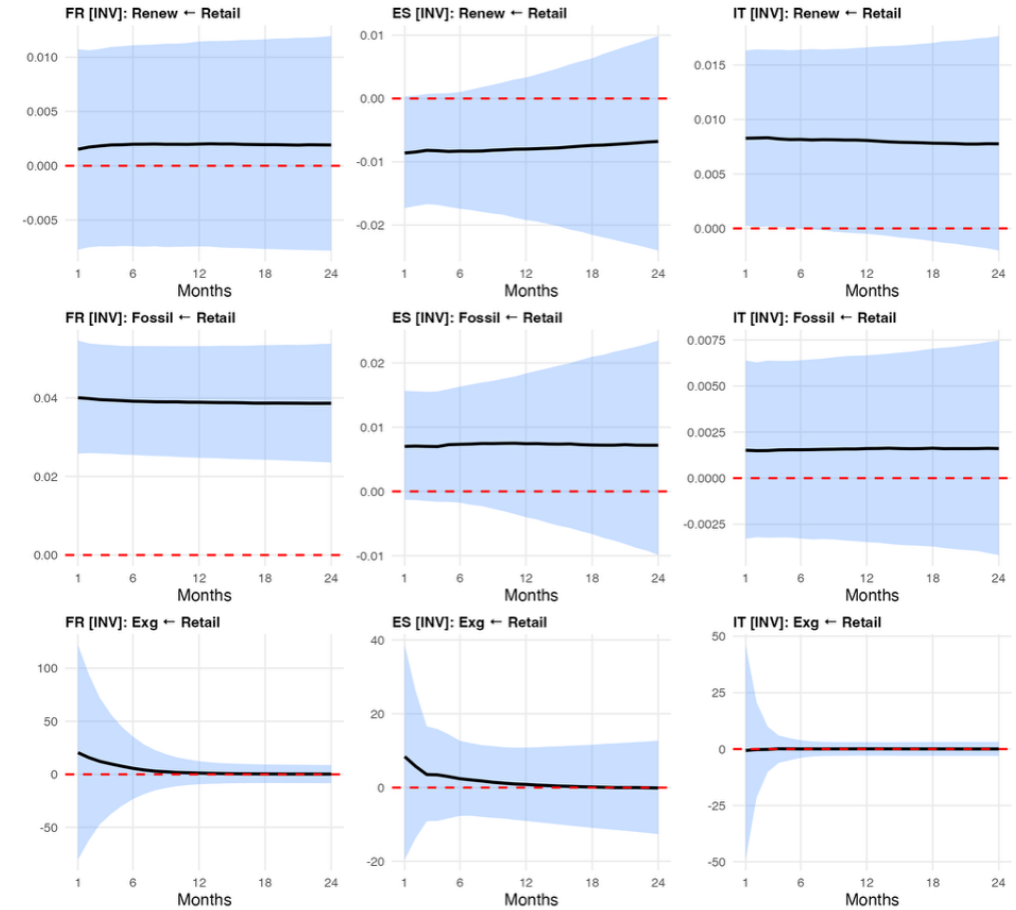


Figure 5: Inverted - Direct Effect of Retail Shock on Energy System

The insignificant of most of these IRFs shows how the consumption channel is not as

strong as the production one in determining energy production mixes.

2. Forecast Error Variance Decomposition

By looking at FEVD, this section examines how the unpredictability of a variable is decomposed in its forecast and assigns to each variable within the full eighteen variable sample the median percentage contribution associated to the short (1 month), medium (12 months) and long (24 months) impact of a standard deviation shock. For both industrial production and retail sales, own shocks retain the highest predictability power by a large margin. Interestingly and aligned with previous analysis, fossil fuel usage is the second largest contributor to the forecast error variance of industrial production for both France and Italy. Retail sales is notably driven by industrial production and the VSTOXX volatility index, embodying the supply side and the uncertainty over future consumption respectively. Interestingly enough, Spanish distinctively rigid labor market makes unemployment rate between the larger explainers of retail sales. On the energy side, the analysis of fossil fuel and renewable energy sources yields consistent results.

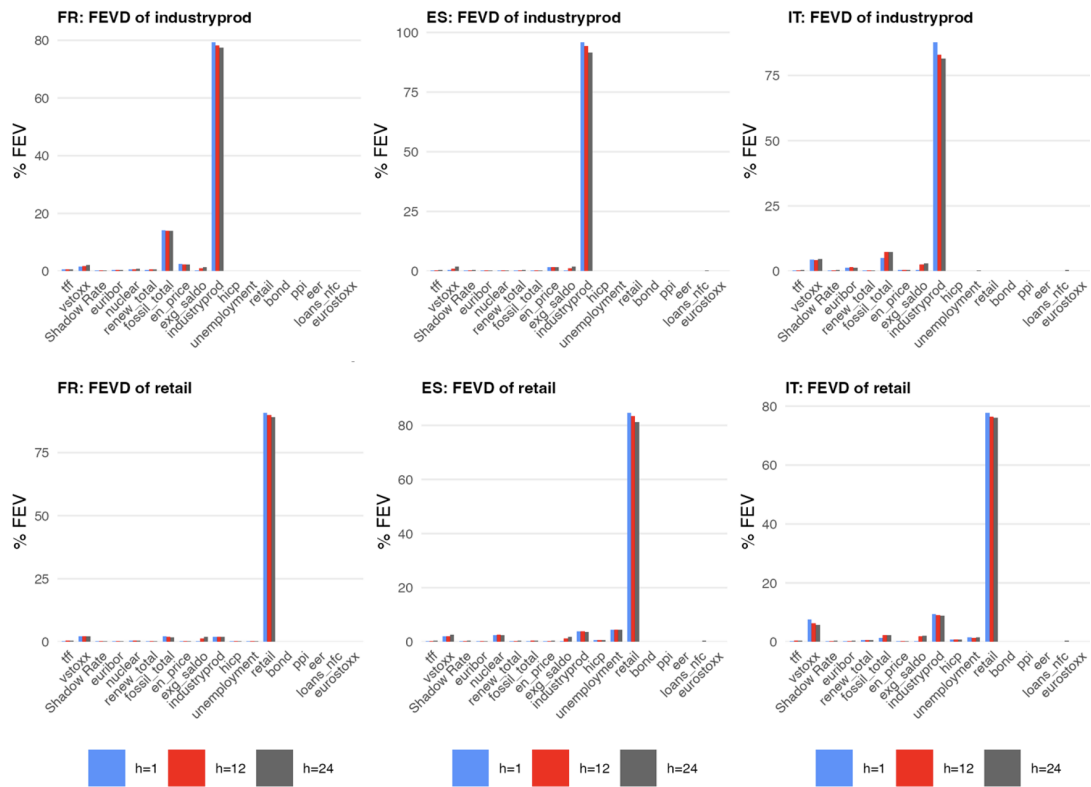


Figure 6: Forecast Error Variance Decomposition

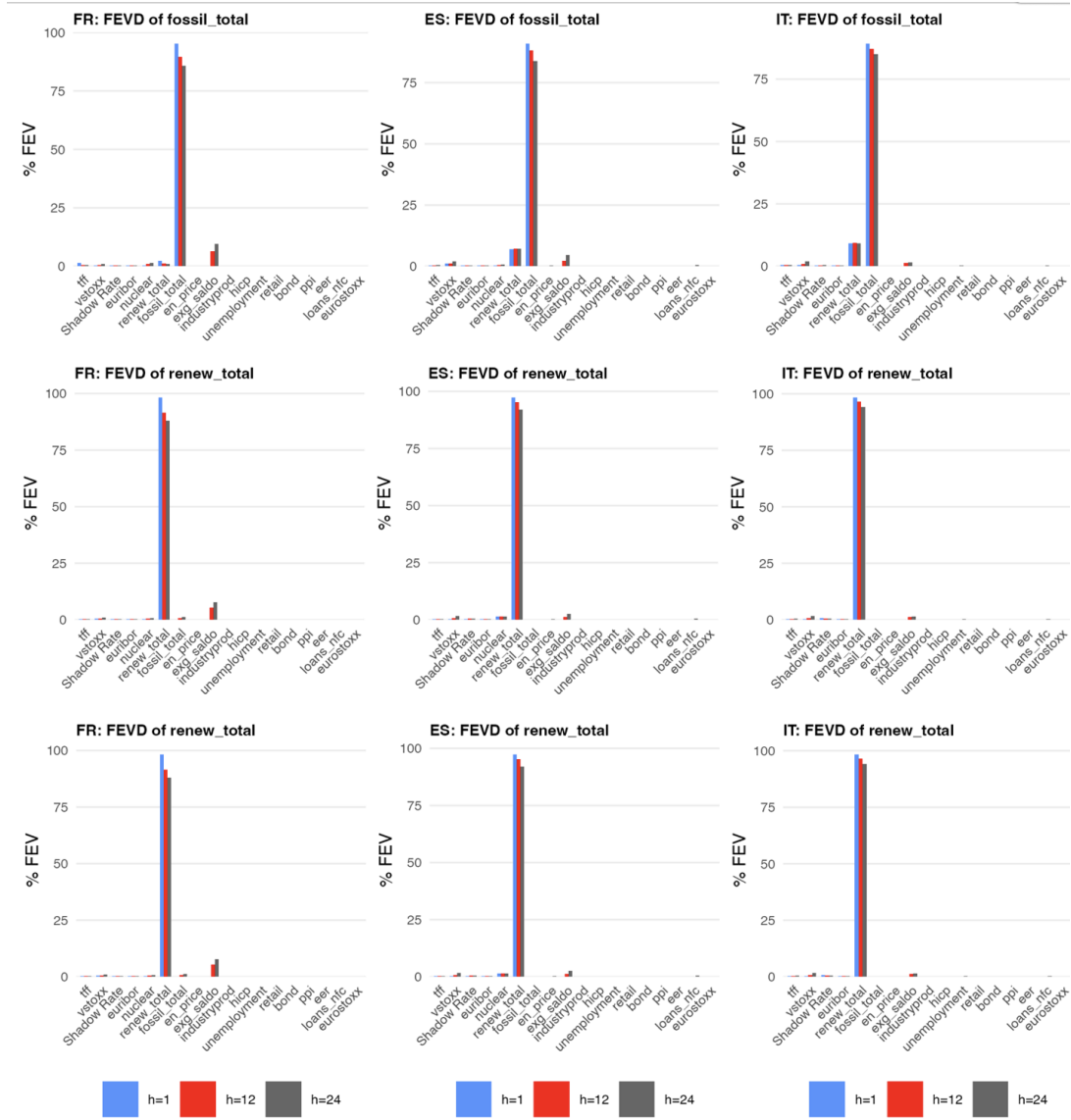


Figure 7: Forecast Error Variance Decomposition 2

3. Time-varying Coefficient BVAR (TVP-BVAR) Impulse Response Functions

To find supporting evidence of the results found for the fixed coefficient estimation and to further investigate the evolution dynamics of the relations that link energy production and macroeconomic variables, this section discusses the results obtained through TVP-BVAR estimation, implemented as specified before. After tuning the hyperparameters, I restrict the variable set used in each model. Keeping $K = 18$ as before would have meant estimating ≈ 25.8 million parameters per model, for a total of over 200 millions stored parameters. To limit this issue and make the computation feasible, I restrict the sample to $K = 5$ for each model, dropping all global variables and some macroeconomic ones, accumulating a total ≈ 2.7 million estimated parameters per model for a total of ≈ 21.6 millions.

The eight estimated models reproduce the mechanics already described in the previous section. The first two models are estimated for all three countries. Both include total fossil fuel generation, total renewable generation, and the wholesale energy price; the first adds the producer price index (PPI) and industrial production, while the second substitutes consumer inflation (HICP) and retail sales. In addition to estimating these models for each of the three countries (yielding six models in total), I estimate two models specific to France. Replacing fossil total with nuclear production, I estimate the producer and consumer channels for nuclear energy as well.

Aligned with the original research question, the Impulse Response Functions are designed to estimate direct and inverse channels for each model, in four different time periods. To inspect the impact of macroeconomic variables on energy production mixes (Hypothesis 1), the first analysis focuses on inverse channels within the sovereign debt crisis (July 2012 corresponds to Draghi’s: “Whatever it Takes”) and Covid 19 pandemic (April 2020 corresponds to most of European countries lockdown imposition). Confronting these results with the 2019 pre-crises baseline (January 2019), the section will end by studying the impact of energy mixes shortages on industry production and retail sales. The direct channel tests the effect of a reduction in energy availability, and its consequent increase in price, on macroeconomic growth and stability (Hypothesis 2). This is done by simulating energy shocks after August 2022, period in which the consequences of the Russian-Ukrainian conflict were underway.

3.1. Inverse Channels

Figure 8 depicts the effects of the inverse channel by displaying the response of energy productions once one standard deviation shock is imposed to industrial production. This shock-response is estimated within each of the three mentioned relevant periods and according to the Cholesky hierarchy imposed it assumes that industry production shock has an impact on PPI, which thereafter influences energy price, and which lastly affects fossil fuels and renewable energies. Exogeneity and chronological order follow the same rules as before.

Figure 9 reproduces these exact dynamics, but it instead focuses on the effects of retail sales within each of these periods. The Cholesky ordering here imposed assumes that retail sales are exogenous and their one standard deviation shock impacts the HICP. The shocked HICP has an effect on energy prices which lastly influences the endogenous variables represented by fossil fuels and renewable energy means of production.

Figure 10 concludes this first analysis by studying the French-specific inverse channels on nuclear energy production. The first panel represents the chained effects of an industrial production shock on nuclear production, the second one the effects of retail sales on it.

Within the first channel, French and Italian dependency on fossil fuels is once again highlighted, especially when considering pre-crisis baseline results of 2019. While these spikes are common for renewable energies within France, Italian increments in industry

production, and thus demands in raw materials and energy sources, do not really impact the amount of renewable energies produced, producing insignificant results until 2020. Insights from Figure 10 suggest that French nuclear production is never really affected by shocks in industrial production. Such a trend confirms the previous hypothesis about a nuclear bulk of energy supply that remains untouched during periods of larger production, with the French economy exploiting fossil and renewable sources whenever demand expands.

Italian reliance on fossil fuels is opposed to Spanish consumption of renewable energies. After the passage of laws granting benefits to zero-emission energy producers (around 2015), Spanish phases of industrial capacity expansion are supported by renewable sources (Ciarreta et al., 2017). These significant changes are visible when comparing pre-bonuses effects on fossil fuels with post-implementation ones. Fossil responses become negative when industrial production shocks are simulated after 2012, renewable energy responses become positive in the long run.

As for the previous analysis, responses through the second inverse channel are less pronounced and explicative. Increases in retail sales and aggregate consumption are, plausibly, less exogenous than industrial production shocks and more credibly trailing patterns in prices. Figures 9 and 10 show however a declining reliance on nuclear sources to meet French consumption needs, mainly replaced by fossil fuels and partially by renewable energies after 2012. The previously observed pattern of resource substitution in Spain is visible through this channel as well but observed only after 2019. The Italian pattern of meeting surges in consumer demand through renewable energy sources is observed only within the 2020 estimates and suggest furthermore that expansions of retail sales are mainly driven by the energy prices and inflation reductions, rather than the other way around.

This hypothesis is furthermore tested in the direct channels analysis with the next IRFs produced being built using the opposite Cholesky ordering and observed during the 2022 energy crisis.

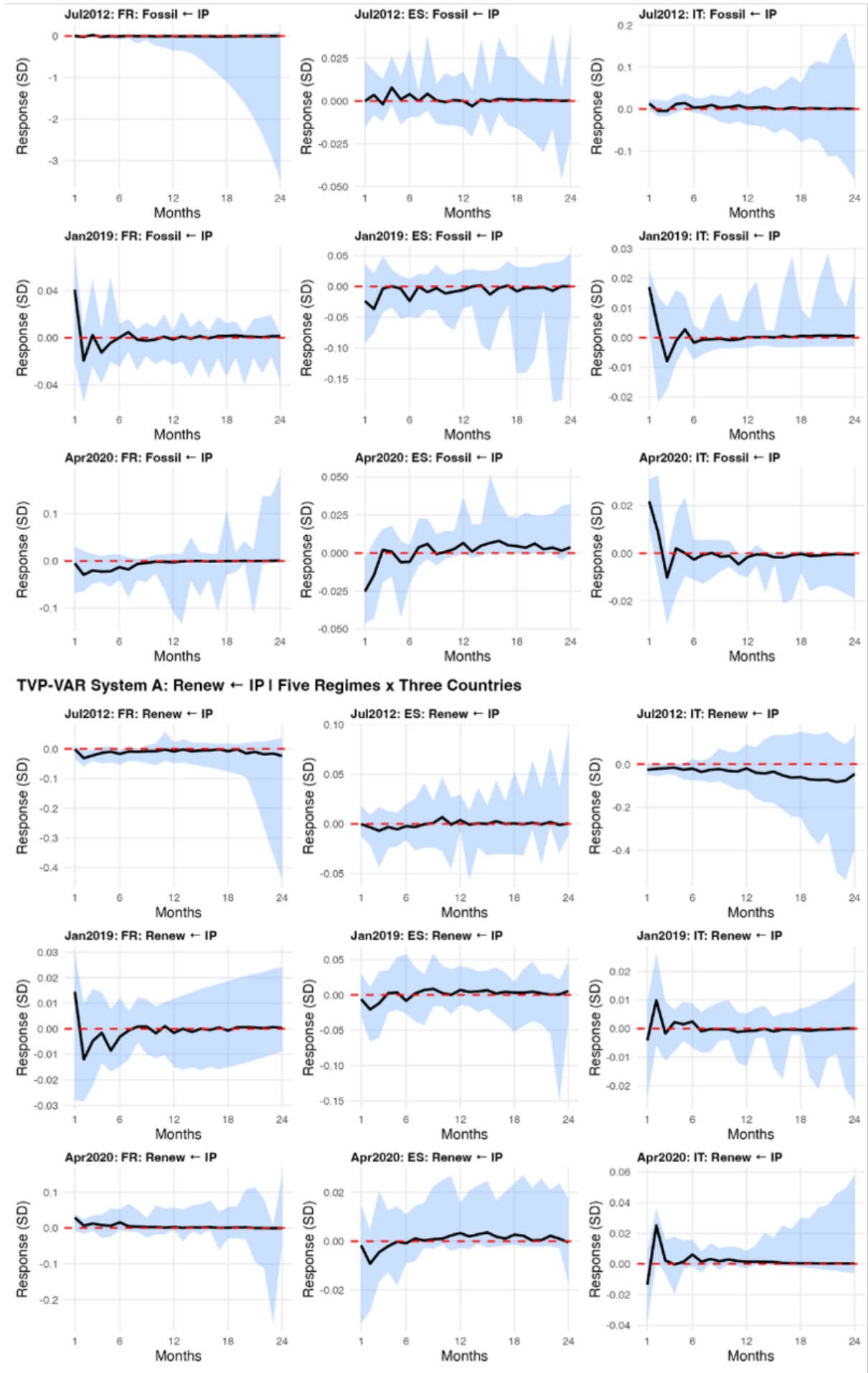


Figure 8: TVP-VAR System A: Fossil / Renew ← IP

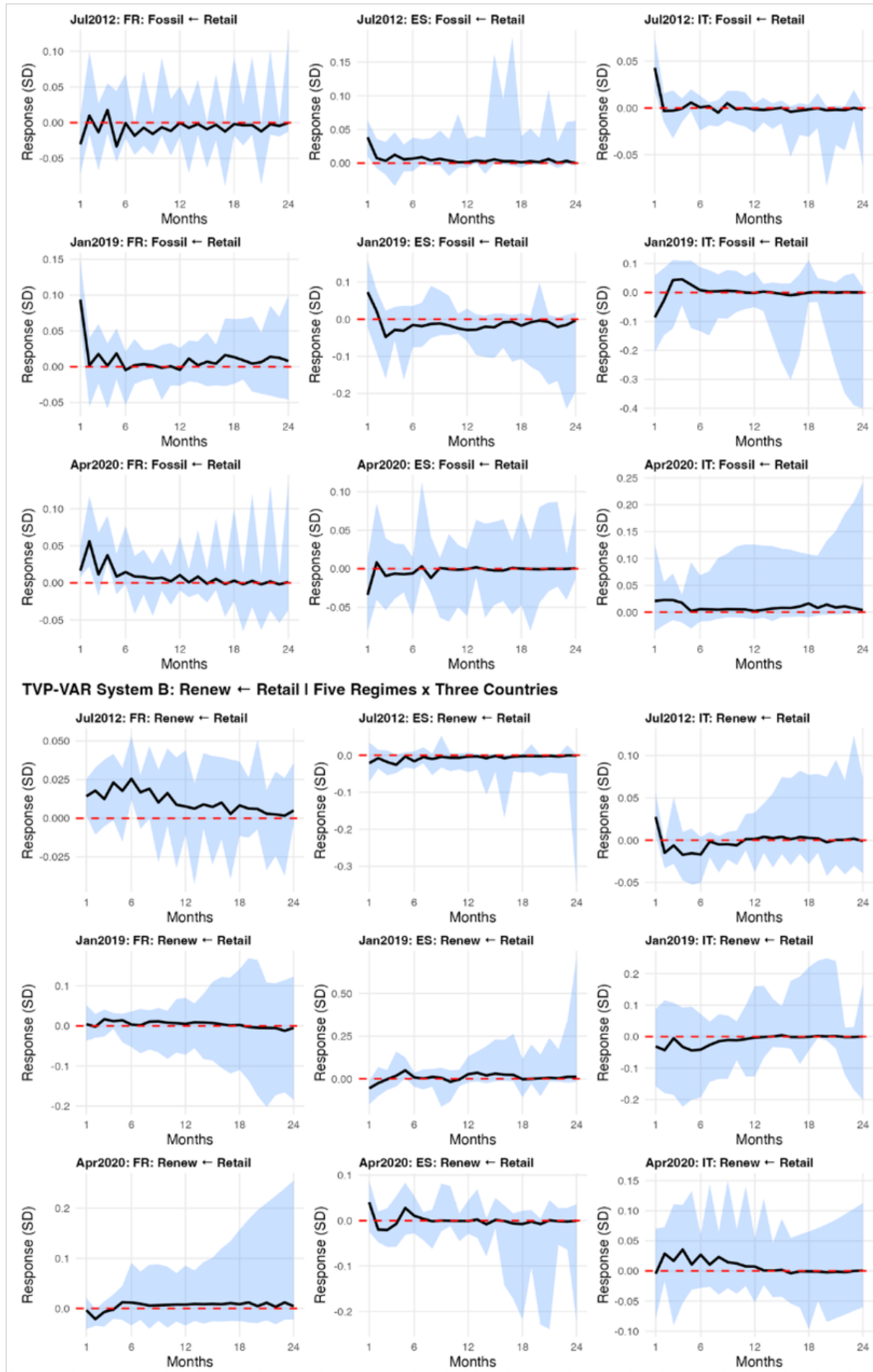


Figure 9: TVP-VAR System B: Fossil / Renew $\leftarrow$ Retail

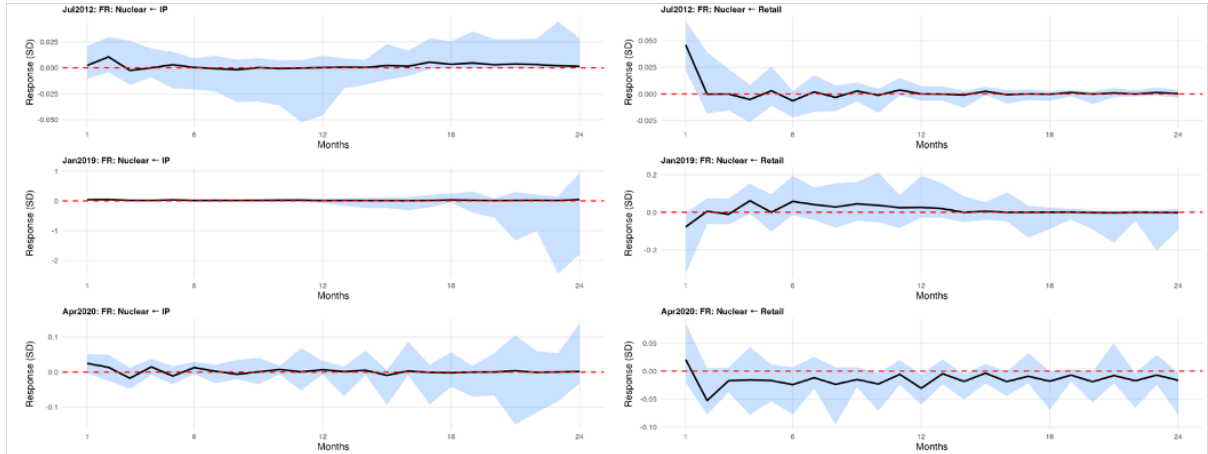


Figure 10: TVP-VAR France Nuclear - System A and B

3.2. Direct Channels

By comparing responses to shocks occurring after the Russian-Ukrainian conflict with those registered in the 2019 baseline, I test the Growth Hypothesis strictly linking energy availability with economic growth. This is done by analyzing the direct channels: shocks to energy supplies that gets transmitted to industry production and retail sales through their two separate channels.

As visible from Figure 11, the unavailability of fossil resources in the aftermath of Russian invasion and sanctions imposed increases the uncertainty of future industrial production particularly for Italy and Spain ([Emiliozzi et al., 2023](#)). The overall effect is estimated to be insignificant and very close to zero but the degree of uncertainty that it carries is evident from the confidence bands evaluated on their respective scales which, compared to 2019, are hundreds of times larger. Once again, Italian reliance on these sources of energy production is visible from the negative responses that industrial production takes on the long run after positive increases in fossil fuel usage, and thus importation. Shortages deriving from the war instantly convert into spikes in energy prices, positively affecting PPI which produces uncertain fluctuations of industrial production.

Such patterns are less visible for countries capable of relying on different sources of production. Spanish uncertainty has a marked positive sign, possibly meaning that the renewable transition has provided a solid base of energy supply and that positive shocks in fossil fuels are associated only with large expansions of industrial production and thus in demand. French smaller uncertainty is driven by the large amount of nuclear energy produced that makes spikes in fossil fuels usage less impactful than for Italy.

As visible from Figure 13, nuclear production within French borders became more unstable in 2022 compared to the baseline. This follows a structural shock occurred in 2022 when a large share of nuclear plants gone through maintenance and stopped their production. The second panel of Figure 11 linking industrial production to renewable energies is characterized by similar insignificant estimates.

For these decomposed estimates as well, direct channels linking energy mixes to

macroeconomic performances are strictly more significant for consumption responses. Figure 12 display these results. The effects of the war are particularly evident on Italian soil, where energy prices and HICP are particularly tied to energy prices. The near-zero baseline response of retail sales to fossil fuel shocks is replaced by a strongly negative response approximately three and a half years later in the aftermath of the conflict. The downward shift of the whole IRF emphasizes the negative effects of the cuts in Russian gas imports and the overall increase in fossil fuel prices. These effects translate into a rising inflation that deteriorates consumers' buying opportunities and their overall consumption (Celasun et al., 2022). Providing evidence for the benefits deriving from energetic independence (Würzburg et al., 2013), the responses of retail sales to shocks in renewable energies acquire a strictly positive sign in 2022. From being uncertain and irrelevant in the long run, positive shocks to sources of energy produced within Italian borders assume a less ambiguous and positive sign after fossil sources availability decreases and their price increase in 2022.

Spanish insignificant estimates carry some insights on the switched signs of the uncertainty produced for fossil energy production across the three years in the sample. The strictly negative uncertainty on the baseline estimate in 2019 is substituted by the positive one in 2022. A potential explanation for this turn is the completion of the renewable transition. While fossil imports in 2019 were a costly alternative for renewable resources still incapable of covering the whole Spanish energetic demand, the 2022 importation is a consequence of an expanding demand and times of positive growth. The second panel point out this relation as well. Renewable shocks in 2019 led to decreases in energy prices and inflation, leading to a larger household consumption, the same shocks in 2022 carried more uncertainty as these sources had become the bulk of the energy supply. Spain stopped to import fossil sources as consequence of renewable shortages and started to import when the economy was flourishing. Renewable sources became fundamental and increases in them started to carry more uncertain outcomes after 2022.

France's estimates are predominantly insignificant, but the large bands carry once again some evidence. French retail sales are positively affected by fossil usage on both periods. United to the large reliance on nuclear energy, these positive responses support a fossil import strategy similar to the Spanish one after 2022. With nuclear able to cover the majority of the country's needs, fossil is imported only when demand for energy increases, thus in periods of economic expansion. On the renewable side, France follows a pattern closer to Italy. With increasing investments in renewable energies representing an uncertain negative impact on consumption before the war, such sources start to become an attractive alternative after the shrinkage of fossil imports and nuclear stoppage.

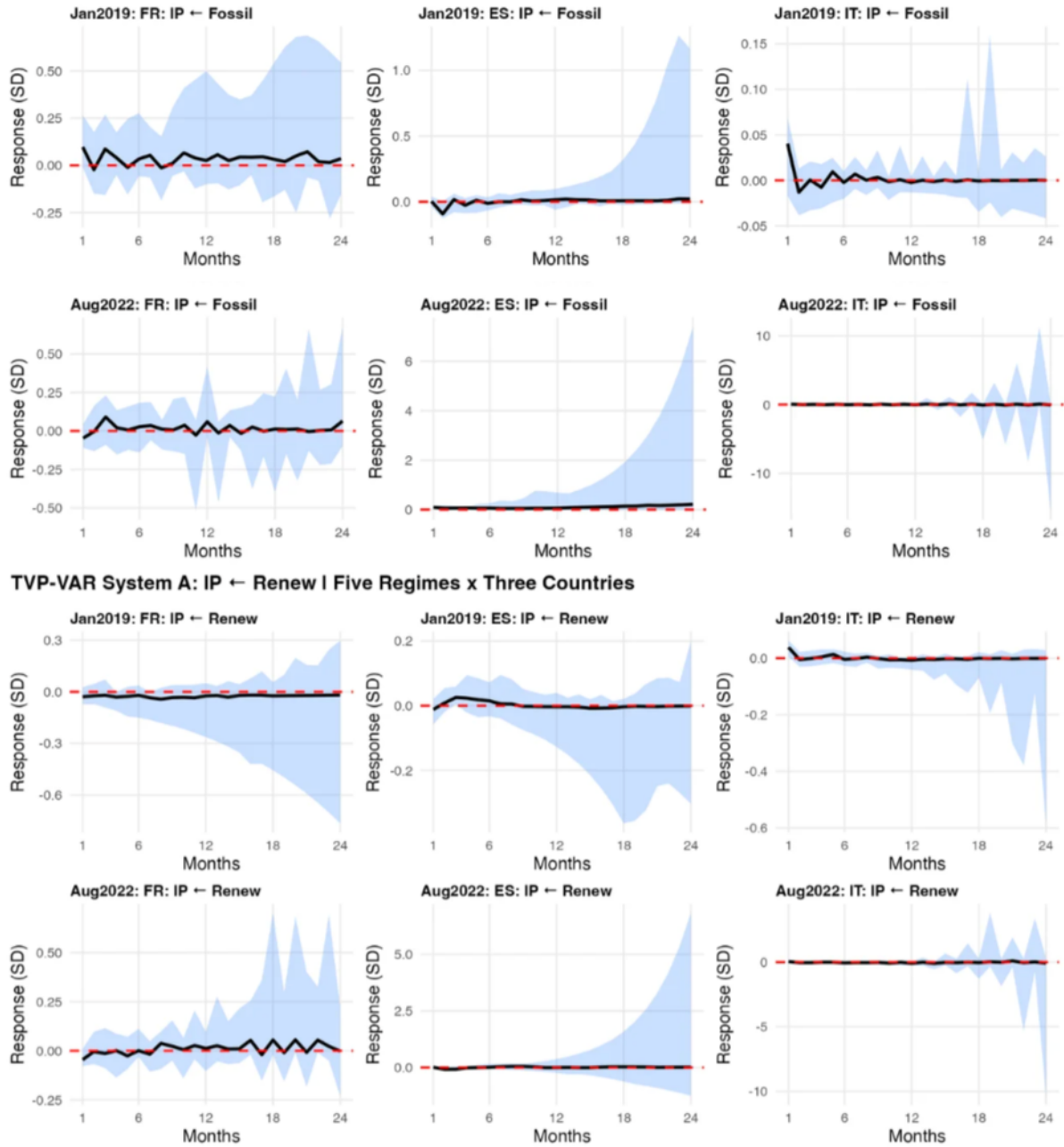


Figure 11: TVP-VAR System A: IP $\leftarrow$ Fossil / Renew

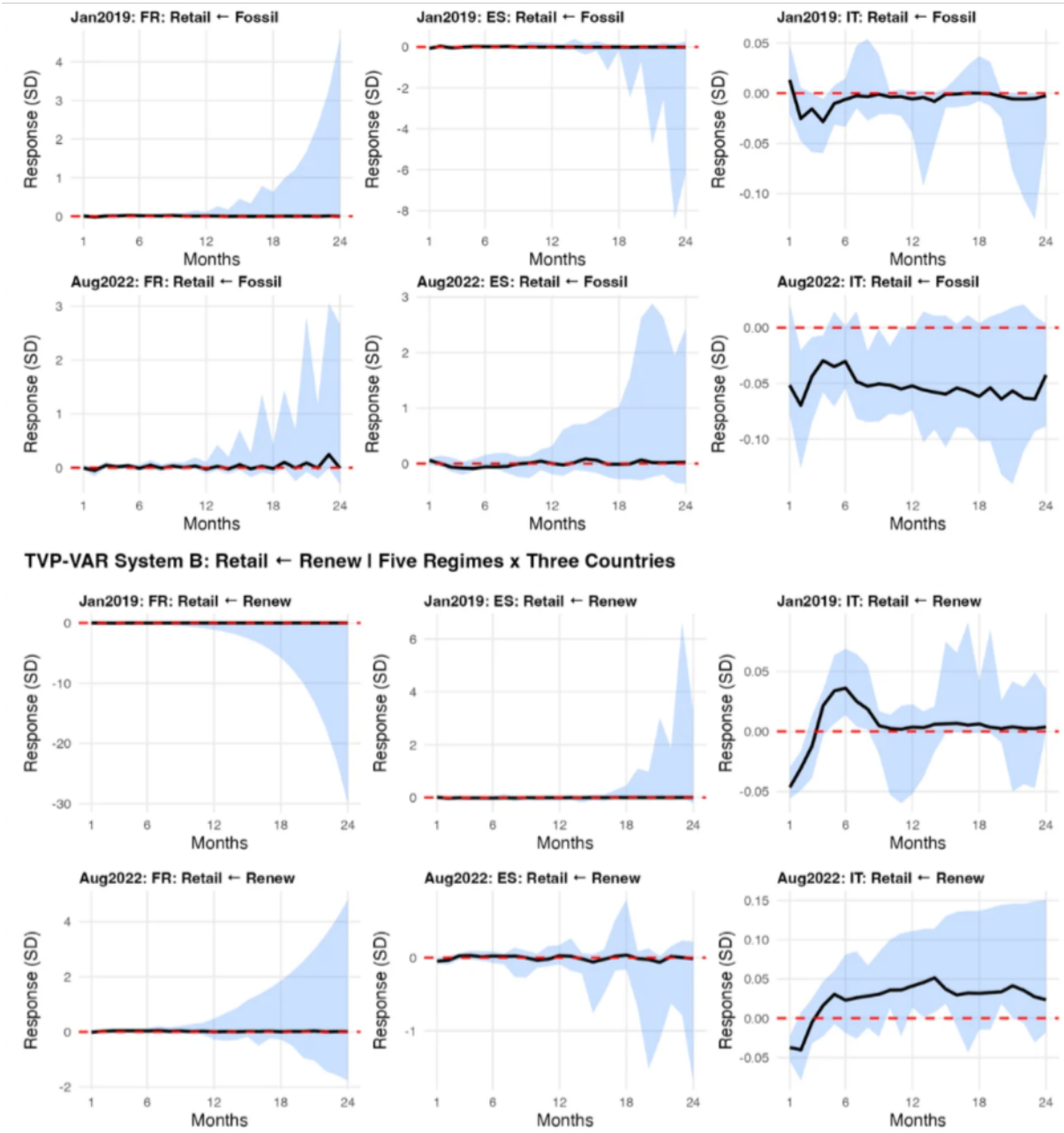


Figure 12: TVP-VAR France Nuclear - System A and B Direct

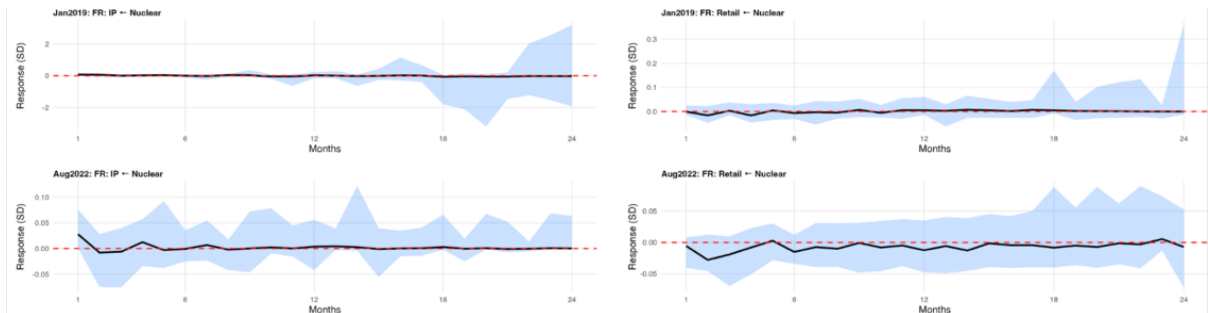


Figure 13: TVP-VAR System B: Retail $\leftarrow$ Fossil / Renew

V. Conclusion

By employing an estimation framework not yet applied in the field of energetic association with macroeconomic performances and through its time-varying extension this paper brings two main contributions.

First, the analysis yielded expected results regarding country-specific patterns of energy dependence. In all sections, it clearly emerged the different composition of energetic production mixes. Italian reliance on fossil fuels has been compared to Spanish renewable mix and French nuclear excellence. These compositional differences were further examined through the time dimension through TVP-IRFs. The analysis highlighted Spanish path towards zero-emission production after 2012, Italian late adoption in the months after Russian-Ukrainian war and French strategies in mixing the three sources.

By comparing these different configurations over time, the study has shown three main advantages of being energetically independent. Apart from the strategic political advantage of not being affected by extra-national events, having access to a broad array of production alternatives allows for the mitigation of effects caused by shortages on any of them and the protection against global trends ([Ang et al., 2015](#)). Just as France can draw on fossil imports or renewable sources during nuclear maintenance periods, Spain can leverage their large solar and wind production at times of international increases of oil prices.

Italy's inability to do so further implies that it cannot capture the economic premiums at times of economic expansion. As proven in previous sections, Italian heavy reliance on imported fossil fuels means that the benefits of positive economic performances are shared with the exporting countries ([Broadstock and Filis, 2014](#)). Spanish and French possibility to reinvest in their territory translates in a more prolonged economic growth and the ultimate reliance on imports whenever the benefits strictly exceed the costs.

The second major finding is that none of the hypothesis formulated in the literature can be uniformly attributed to all economic outcomes. Empirical evidence suggests that the relationship between energy generation and macroeconomic performance depends critically on the specific transmission channel through which it operates. Results from the impulse response functions reveal a concrete and measurable feedback effect existing within the first channel, being not significant for the second one. While sources of energetic production impact industry production as well as industry production influences them, the inverse channel of household consumption is found not to exert significant effects on energetic mixes. Reasonably enough, supply-side shocks attributable to fiscal policy or technological change exert a greater influence on energy demand than what exogenous spikes in consumption do.

While the imposed Cholesky ordering for the inverse production channel has some economic meaningful sense (improvement in industry production may negatively affect PPI and the subsequent chain), the inverse consumption channel rests on less defensible

identifying assumptions (retail sales and consumption notoriously trail HICP movements, the opposite being unlikely). This is further supported by results in the FEVD decomposition that display significant effects of uncertainty volatility indexes and of other macro-variable, such as employment, on retail sales but not on industry production. The consistency of significant results across channels allows us to discard the fourth hypothesis of null interactions, the decomposition of the economy into supply- and demand-side channels represents a novel contribution that evaluates the predominancy of feedback effects on the supply side and the existence of a stronger effect of energy generation on household consumption than the vice-versa (growth hypothesis 2).

In conclusion, the study has shown positive effects of energetic independence by studying the evolution of shocks over time and across countries. It has additionally evaluated previously established hypotheses motivating the analysis of macroeconomic outcomes through distinct empirical frameworks. Future research should focus on determining how each separate energy production source impact both channels and how economic outcomes interact with them. Another potential avenue could investigate and use these models to predict future performances of each country by imposing shocks or by changing energy generations. In a future exercise it could be interesting to describe how each country would have reacted or will react in the future if they had neighboring production mixes. Understanding how the Italian economy would have fared during the 2012 debt crisis if it had a solid nuclear energy production or how would France have responded to the Russian conflict if it had Spanish renewable capacities are all policy-relevant questions for which all the coefficients have already been estimated in this paper. A final valuable addition would involve the recovering of French wholesale trade prices for the whole sample. While producing more reliable results for all estimates, such dataset improvement would also correct for the biggest limitation of this study. BVAR and TVP-BVAR estimates do not account for the uncertainty deriving from the initial imputation exercise, treating them as all other observations. This could potentially hinder the validity of the results and make the cross-country comparisons flawed.

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Appendix A.

Country	Variable	Adjustment Status
FR	industryprod	SEASONALITY DETECTED → adjusted
FR	industryprod_manif	SEASONALITY DETECTED → adjusted
FR	industryprod_elect	SEASONALITY DETECTED → adjusted
FR	retail	SEASONALITY DETECTED → adjusted
FR	bop	SEASONALITY DETECTED → adjusted
FR	hydro	SEASONALITY DETECTED → adjusted
FR	nuclear	SEASONALITY DETECTED → adjusted
FR	fossil_total	SEASONALITY DETECTED → adjusted
FR	wind	SEASONALITY DETECTED → adjusted
FR	solar	SEASONALITY DETECTED → adjusted
FR	gas	SEASONALITY DETECTED → adjusted
FR	renew_total	SEASONALITY DETECTED → adjusted
FR	biomass	no seasonality or X-13 failed → kept original
FR	renew_other	no seasonality or X-13 failed → kept original
ES	industryprod	SEASONALITY DETECTED → adjusted
ES	industryprod_manif	SEASONALITY DETECTED → adjusted
ES	industryprod_elect	SEASONALITY DETECTED → adjusted
ES	retail	SEASONALITY DETECTED → adjusted
ES	bop	SEASONALITY DETECTED → adjusted
ES	hydro	SEASONALITY DETECTED → adjusted
ES	nuclear	SEASONALITY DETECTED → adjusted
ES	fossil_total	SEASONALITY DETECTED → adjusted
ES	wind	SEASONALITY DETECTED → adjusted
ES	solar	SEASONALITY DETECTED → adjusted
ES	gas	SEASONALITY DETECTED → adjusted
ES	renew_total	SEASONALITY DETECTED → adjusted
ES	renew_other	SEASONALITY DETECTED → adjusted
ES	biomass	no seasonality or X-13 failed → kept original
IT	industryprod	SEASONALITY DETECTED → adjusted
IT	industryprod_manif	SEASONALITY DETECTED → adjusted
IT	industryprod_elect	SEASONALITY DETECTED → adjusted
IT	retail	SEASONALITY DETECTED → adjusted
IT	bop	SEASONALITY DETECTED → adjusted
IT	hydro	SEASONALITY DETECTED → adjusted
IT	fossil_total	SEASONALITY DETECTED → adjusted
IT	wind	SEASONALITY DETECTED → adjusted
IT	solar	SEASONALITY DETECTED → adjusted
IT	gas	SEASONALITY DETECTED → adjusted
IT	renew_total	SEASONALITY DETECTED → adjusted
IT	biomass	no seasonality or X-13 failed → kept original
IT	geothermal	no seasonality or X-13 failed → kept original
IT	renew_other	no seasonality or X-13 failed → kept original

Table A.1: X-13 Seasonal Filter Adjustment Status

Notes: The filtering procedure follows [Sax and Eddelbuettel \(2018\)](#) implementation.

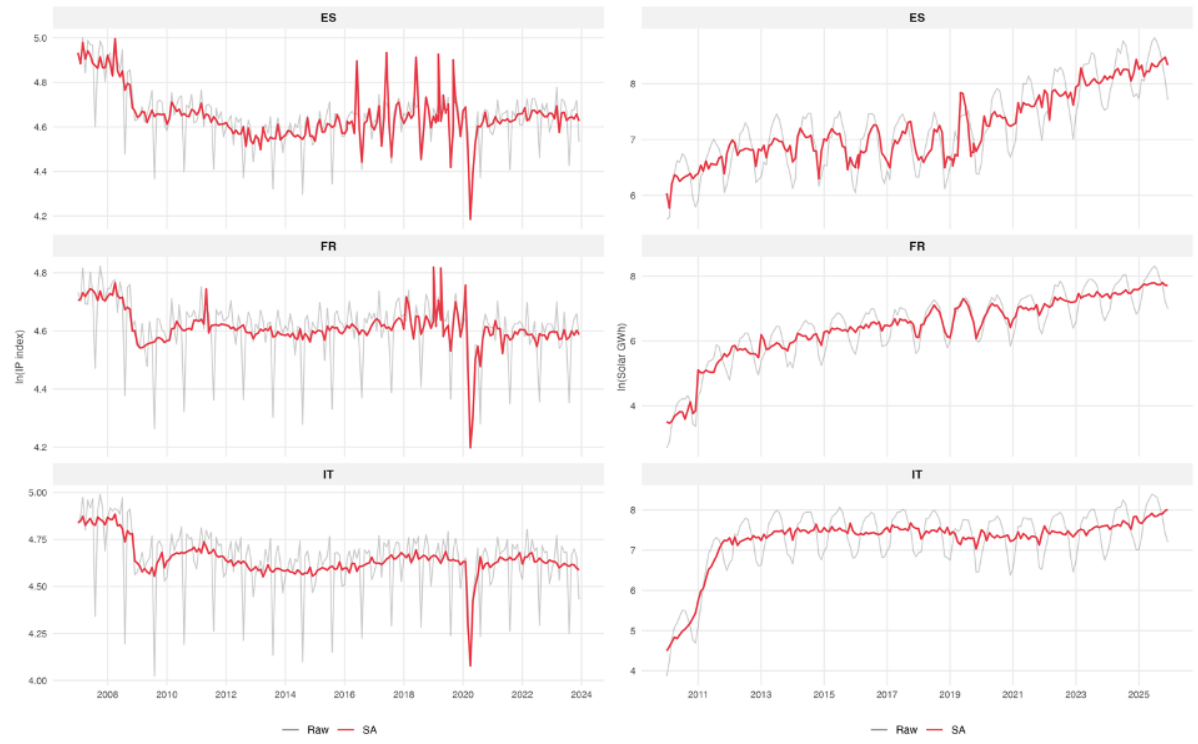


Figure A.1: Seasonality Transformed Series: Industry Production and Solar Generation.

Notes: The figure shows the comparison between raw values (grey) and the seasonally adjusted ones (red) for two example variables. Industrial Production and Solar Generation suffer from similar cyclical trends among the chosen countries. These patterns were corrected by the X-13 seasonal filter. Interestingly and effectively, the uncommon nature of the shocks caused by Covid-19 pandemic were not corrected by the filter.

Appendix B.

Variable	Test Type	Statistic	Lags (AIC)	1% CV	5% CV	10% CV	Conclusion
tff	drift	−1.560	11	−3.46	−2.88	−2.57	I(1)
vstxxx	drift	−3.898	2	−3.46	−2.88	−2.57	I(0)
Shadow Rate	trend	−0.794	4	−3.99	−3.43	−3.13	I(1)
euribor	trend	−2.649	12	−3.99	−3.43	−3.13	I(1)
nuclear	drift	−1.877	12	−3.46	−2.88	−2.57	I(1)
renew_total	trend	−1.988	2	−3.99	−3.43	−3.13	I(1)
fossil_total	drift	−3.677	1	−3.46	−2.88	−2.57	I(0)
en_price	drift	−2.780	12	−3.46	−2.88	−2.57	I(1)*
exg_saldo	none	−1.781	2	−2.58	−1.95	−1.62	I(1)*
industryprod	trend	−5.087	2	−3.99	−3.43	−3.13	I(0)†
hicp	trend	−3.347	12	−3.99	−3.43	−3.13	I(1)*
unemployment	drift	−1.656	12	−3.46	−2.88	−2.57	I(1)
retail	trend	−5.643	1	−3.99	−3.43	−3.13	I(0)†
bond	trend	−0.613	3	−3.99	−3.43	−3.13	I(1)
ppi	drift	−0.534	5	−3.46	−2.88	−2.57	I(1)
eer	drift	−1.909	1	−3.46	−2.88	−2.57	I(1)
loans_nfc	trend	−3.611	12	−3.99	−3.43	−3.13	I(0)
eurostoxx	drift	−2.567	7	−3.46	−2.88	−2.57	I(1)

Table B.1: Augmented Dickey-Fuller (ADF) Unit Root Test Results

Notes: The conclusion $I(0)$ or $I(1)$ is determined using the strict 5% critical value threshold. Maximum lag length was set to 12, with optimal lags selected via the Akaike Information Criterion (AIC).

(*) *Borderline Stationarity:* While classified as $I(1)$ at the 5% level, the null hypothesis of a unit root is formally rejected at the 10% level for wholesale electricity prices (*en_price*), net exchange balance (*exg_saldo*), and consumer inflation (*hicp*).

(†) *Structural Break Distortion:* Macroeconomic aggregates such as *retail* and *industryprod* are structurally $I(1)$, but reject the null hypothesis of a unit root in this sample. This is the mentioned limitation of linear ADF tests, which tend to misinterpret massive, deterministic structural breaks as highly mean-reverting stationary processes.

Table B.2: Johansen Cointegration Trace Test: Italian Retail Sales and HICP

Null Hypothesis	Trace Statistic	10% CV	5% CV	1% CV
$r = 0$	28.20	17.85	19.96	24.60
$r \leq 1$	8.17	7.52	9.24	12.97

Notes: The test was conducted using a Trace statistic with 4 lags and a constant. The null hypothesis of no cointegration ($r = 0$) is rejected at the 5% level, while the null of at most one cointegrating relationship ($r \leq 1$) cannot be rejected. This confirms exactly one formal long-run equilibrium ($r = 1$) between the indices.

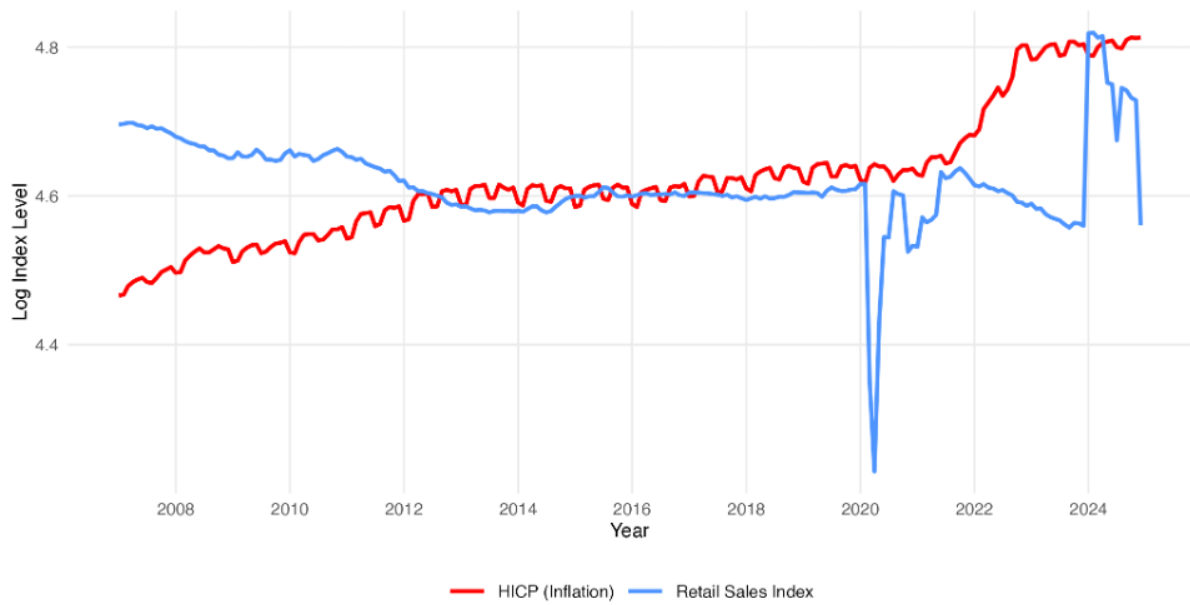


Figure B.1: Italian Retail Sales and HICP Cointegrated Variables

Appendix C.

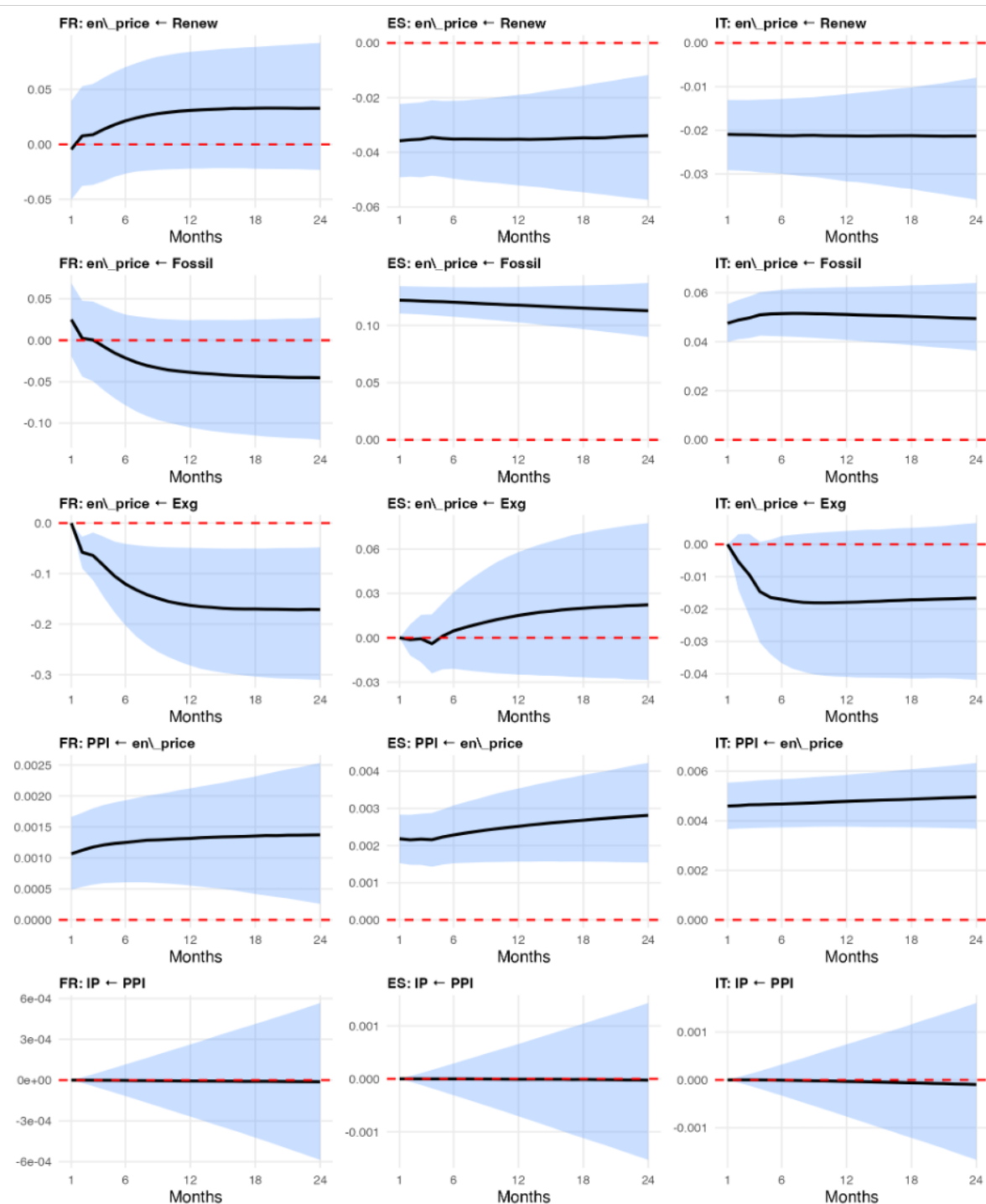


Figure C.1: Channel 1 - Complete Chain Decomposition

Notes: The decomposed first direct transmission channel defined exhibits the impact of each energy variable on the price, the subsequent effect of the price on the PPI and how does PPI influences IP. The main result of this analysis is the negative impact on energy prices by renewable energies, true both for Italy and Spain, coupled with the positive effect that fossil have. Independent sources of production like

renewables are interestingly shifting the prices down, with a decrease of 4 p.p and 2 p.p. for Spain (14% one-standard deviation increase) and Italy (12% one-standard deviation increase) respectively.

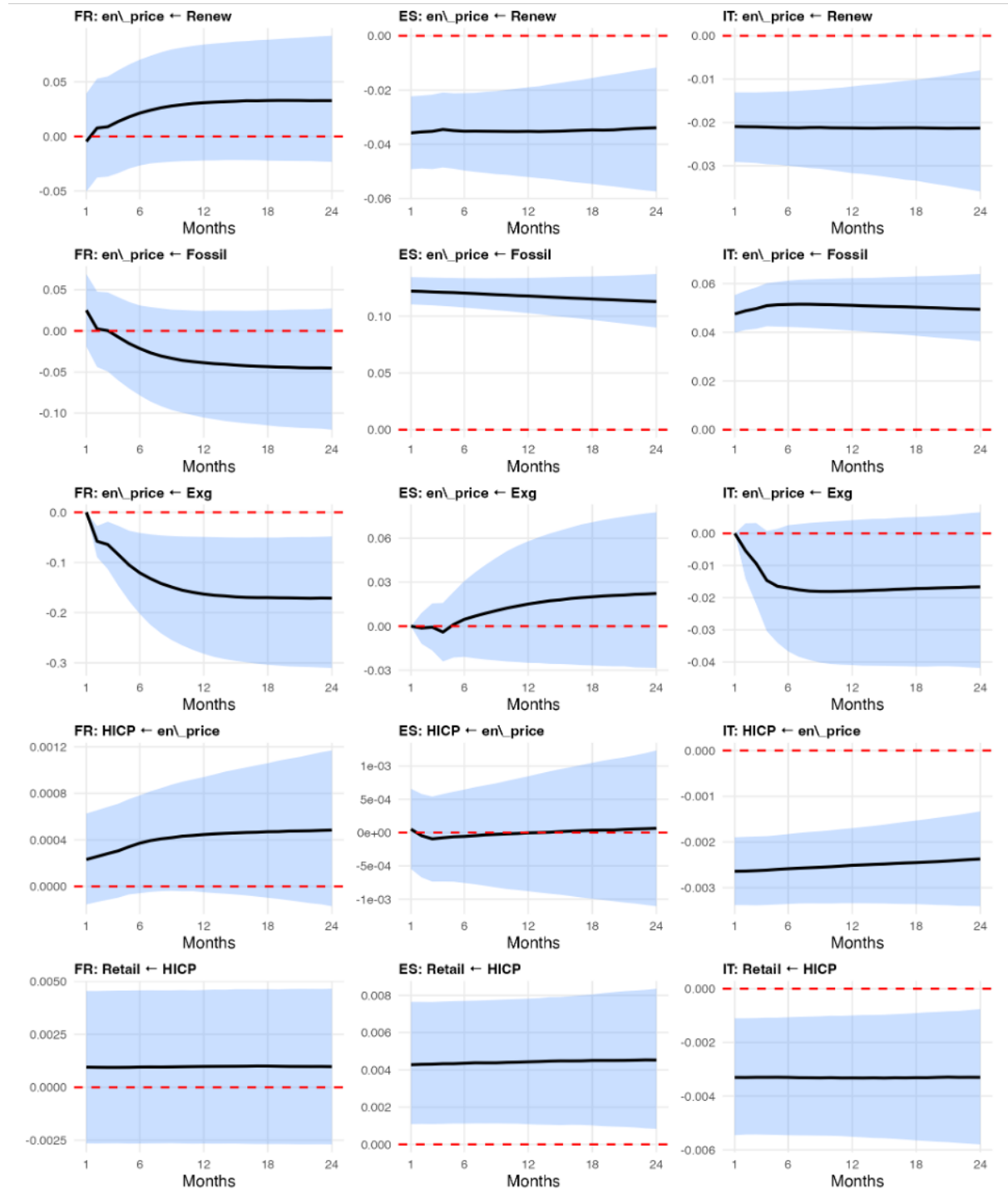


Figure C.2: Channel 2 - Complete Chain Decomposition

Notes: The decomposed second direct transmission channel exhibits the impact of each energy variable on the price, the subsequent effect of the price on the HICP and how does HICP influences retail sales.

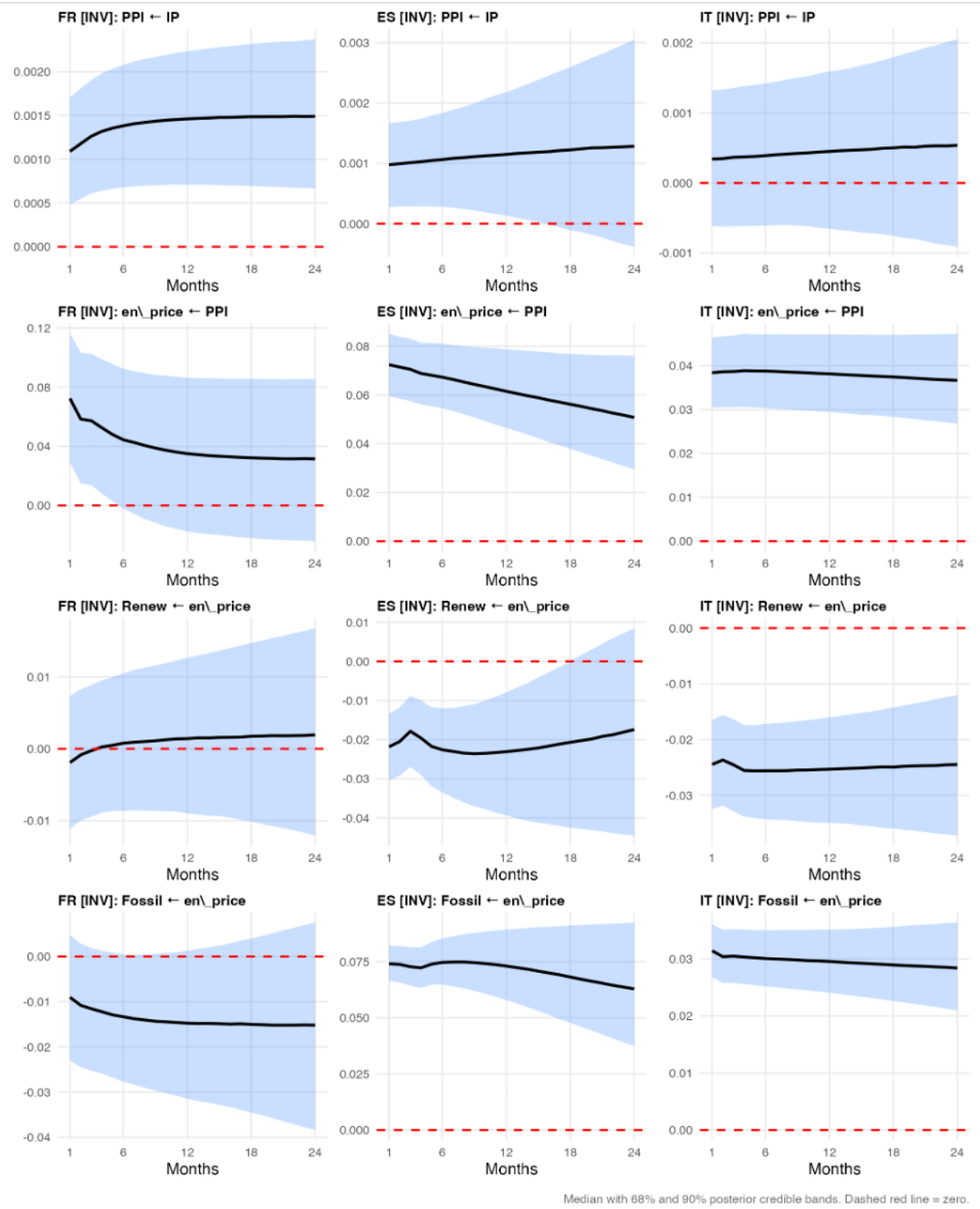


Figure C.3: Inverted Channel 1 - Complete Chain Decomposition

Notes: The decomposed first inverse channel exhibits the impact of IP on PPI, its subsequent effect on energy prices and the interaction of them with each production mix. It further captures the negative relation existing between renewable sources of energy and the prices.

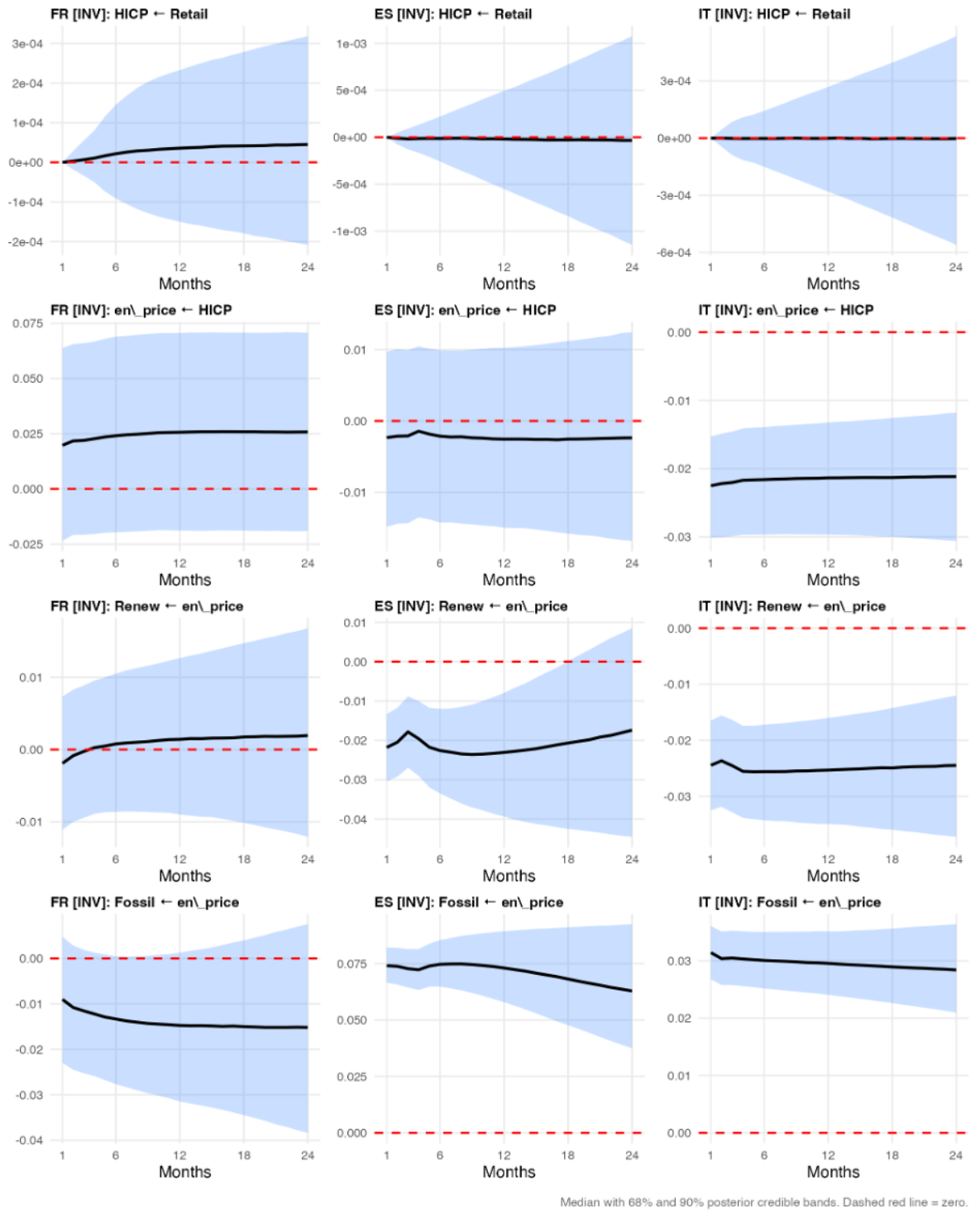


Figure C.4: Inverted Channel 2 - Complete Chain Decomposition

Notes: While presenting the same effects of energy prices on the different sources of energy production, this decomposed table show the first two chained transmissions. As visible, retail sales have insignificant impact on HICP, whose effect on energy prices is ambiguous and difficult to interpret. As explained before, these responses are likely to be inversely specified and thus impossible to interpret.